



**COMSATS University Islamabad, Vehari
Campus**

Software Requirements Specification

(SRS Document)

for

**FYP Proposal Approval System (for CUI
Vehari)**

Version 4.0

By

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Revision History

Author(s)	Date	Change Description	Version
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Comments (by Committee)	Action Taken
*Include the ones given at scope time both in doc and presentation	

Signature: _____

Abstract

This project presents the development of a Final Year Project (FYP) Proposal Approval System for COMSATS University Islamabad, Vehari Campus. The current manual, paper-based process for managing student projects suffers from slow approval workflows, a lack of transparency, difficulty in finding suitable supervisors, and inefficient communication between stakeholders.

We can define the core problem as follows: the traditional process creates bottlenecks that delay project approvals by weeks and requires excessive manual follow-up for status updates. This project solves these problems by creating a web-based platform that automates the entire FYP management process from initial idea submission to final defence scheduling.

The system supports four types of users: Students, Supervisors, Evaluation Committee Members, and Administrators. Students can submit project ideas and track their status in real-time. Supervisors can publish their research interests, review submissions, and manage their project workload. Administrators can manage users, form evaluation committees, and schedule defence sessions. The system architecture uses the Laravel 12 framework with PHP 8.0 or later and MySQL database, following the Model-View-Controller (MVC) pattern.

Key features include role-based access control to ensure users only access authorized features, document management with version control for scope documents, and an intelligent scheduling system that prevents conflicts. The system can handle at least 200 concurrent users with page load times under 3 seconds. For security, all passwords are encrypted using the bcrypt algorithm, and the system protects against common web attacks like SQL injection and cross-site scripting.

Testing involved unit testing, integration testing, performance testing with simulated concurrent users, security testing, and user acceptance testing with actual students and faculty members. The expected results show that processing time for project approvals will reduce by approximately 60–70%, from several weeks to just a few days. The system provides real-time status tracking that eliminates the need for manual communication overhead by 70%.

This project demonstrates how modern web technologies can solve real administrative challenges in educational institutions. The system is designed to be maintainable and scalable, with potential future enhancements including project progress monitoring, milestone tracking, and plagiarism detection integration.

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1 Introduction

1.1 Project Scope

The FYP Proposal Approval System aims to digitise and streamline the management of student projects at COMSATS University Islamabad, Vehari Campus. The system will facilitate project idea submission, supervisor allocation, scope document management, and defence scheduling through an integrated web-based platform.

Key features include:

- User management with role-based access control
- Supervisor profile and research interest publication
- Project idea submission and approval workflow
- Scope document submission and version management
- Evaluation committee management and defence scheduling
- Administrative dashboard with analytics and reporting

Out of Scope:

- Integration with the CUI central online system
- Mobile application development
- AI-powered FYP idea suggestion system
- Payment processing or financial management
- External stakeholder management (industry partners)

1.2 Project Significance

The development of this FYP Proposal Approval System addresses several critical challenges in the current manual process at COMSATS University Islamabad, Vehari Campus:

Current Challenges:

- Paper-based submission processes leading to delays and inefficiencies
- Lack of transparency in the approval workflow
- Difficulty in tracking project status and deadlines
- Manual committee formation and scheduling conflicts
- Limited accessibility to the supervisor's research interests

Expected Benefits:

- Efficiency: Reduced processing time for project approvals from weeks to days
- Transparency: Real-time status tracking for all stakeholders
- Accessibility: 24/7 access to the system from anywhere with internet connectivity
- Organization: Centralized repository for all project-related documents
- Analytics: Data-driven insights for administrative decision-making
- Standardization: Consistent evaluation criteria and documentation formats

This system will serve as a foundational platform that can be extended in future phases to include additional features such as project monitoring, milestone tracking, and alumni project archives.

1.3 Target Audience

This Software Requirements Specification (SRS) document is intended for the following stakeholders:

Primary Audience:

- Development Team: Software developers, database administrators, and UI/UX designers who will implement the system
- Project Supervisor: Faculty member overseeing the development project
- Quality Assurance Team: Testers who will validate the system against these requirements

Secondary Audience:

- FYP Committee: Administrative staff who will use and maintain the system
- Department Head: Decision-maker for system approval and deployment
- Future Developers: Individuals who may maintain or enhance the system
- System Administrators: IT staff responsible for deployment and infrastructure

End Users (not primary audience for SRS, but beneficiaries):

- Students (Final Year)
- Supervisors (Faculty Members)
- Evaluation Committee Members
- Administrative Staff

1.4 Project Overview

The FYP Proposal Approval System is a comprehensive web-based application designed to replace the traditional paper-based final year project management process at COMSATS University Islamabad, Vehari Campus.

System Architecture: The system will be built using modern web technologies with a three-tier architecture:

- Presentation Layer: Responsive web interface built with Laravel Blade templates, Tailwind CSS
- Application Layer: Backend logic implemented using Laravel 12 framework with PHP 8.0 or later
- Data Layer: MySQL relational database for persistent storage

Key Modules:

1. User Management: Authentication, authorization, and profile management
2. Supervisor Profile Management: Research interests and capacity tracking
3. Project Idea and Approval: Submission workflow and status management
4. Scope Document Management: Document upload, review, and version control
5. Evaluation Committee Management: Committee formation and defence scheduling

System Workflow Overview:

1. Supervisors publish research interests and available slots
2. Students browse supervisors and submit project ideas
3. Supervisors review and approve/reject/request revisions
4. Students upload scope documents for approved ideas
5. Supervisors review and approve scope documents
6. Administrators form evaluation committees and schedule defences
7. Committee members evaluate projects and submit feedback
8. System generates reports and maintains project history

Technology Stack:

- Framework: Laravel 12 (PHP 8.0 or later)
- Frontend: Tailwind CSS / Blade template
- Database: MySQL
- Authentication: Laravel Sanctum
- File Storage: Local storage with PDF support

Development Timeline: The project will be developed in phases according to the Gantt chart provided in the appendix, with key milestones including requirements analysis, design, implementation, testing, and deployment.

2 Software Requirement Specification

2.1 Software Requirements Technique

Requirements Elicitation Methods

2.1.1 Stakeholder Interviews

- Conducted interviews with FYP committee members, faculty supervisors, and students
- Identified pain points in the current manual process
- Gathered expectations and desired features from different user perspectives

2.1.2 Document Analysis

- Reviewed existing FYP guidelines and policies from COMSATS University Islamabad
- Analyzed sample project proposal forms and scope documents
- Studied the current approval workflow and timeline requirements

2.1.3 Observational Studies

- Observed the current paper-based submission and approval process
- Documented bottlenecks and inefficiencies in the existing system
- Identified opportunities for automation and improvement

2.1.4 Use Case Analysis

- Developed use case diagrams to model system interactions
- Created detailed use case descriptions for key functionalities
- Validated use cases with stakeholders for completeness

2.1.5 Prototyping

- Created low-fidelity wireframes for user interface concepts
- Gathered feedback on proposed workflows and screen designs
- Iteratively refined requirements based on prototype feedback

2.2 Functional Requirements

2.2.1 User Management Module

FR-1: User Registration

Table 2.1: FR-1: User Registration

Field	Details
Requirement ID	FR-1
Title	User Registration
Requirement	The system shall allow the Administrator to upload a CSV file containing user data to create multiple user accounts at once.
Source	Project Proposal
Rationale	To efficiently create accounts for large numbers of users without manual entry.
Business Rule	The CSV file must contain the required fields in the specified format.
Dependencies	None
Priority	High

Table 2.1 shows how administrators can upload CSV files to create multiple user accounts at once. This is really useful when you need to add lots of students and faculty at the start of each semester.

FR-2: Role-Based Access Control

Table 2.2: FR-2: Role-Based Access Control

Field	Details
Requirement ID	FR-2
Requirement	The system shall restrict access to features based on user roles, with each role having predefined permissions.
Source	Project Proposal
Rationale	To ensure that users can only access features relevant to their role.
Business Rule	A user can be assigned multiple roles if necessary.
Dependencies	FR-1
Priority	High

Table 2.2 defines the role-based access control to make sure users only see what they're supposed to. Students can't access admin functions and vice versa.

FR-3: User Authentication

Table 2.3: FR-3: User Authentication

Field	Details
Requirement ID	FR-3
Requirement	The system shall authenticate users through email and password, with password reset functionality via email.
Source	Project Proposal
Rationale	To secure access to the system and verify user identity.
Business Rule	Passwords must meet minimum security requirements.

Dependencies FR-1

Priority High

Table 2.3 specifies the login system with email and password authentication. It also includes password reset functionality for when people forget their credentials.

FR-4: User Profile Management

Table 2.4: FR-4: User Profile Management

Field	Details
Requirement ID	FR-4
Requirement	The system shall allow users to update their profile information and change their password.
Source	Project Proposal
Rationale	To maintain accurate user information and account security.
Business Rule	Email address changes require verification.
Dependencies	FR-1, FR-3
Priority	Medium

Table 2.4 allows users to update their profile info and change passwords. Email changes need verification for security reasons.

FR-5: User Deactivation

Table 2.5: FR-5: User Deactivation

Field	Details
Requirement ID	FR-5

Requirement	The system shall allow the Administrator to deactivate user accounts while preserving historical data.
Source	Project Proposal
Rationale	To manage system access for users who have left the institution.
Business Rule	User accounts can be reactivated if needed.
Dependencies	FR-1
Priority	Medium

Table 2.5 lets administrators deactivate accounts when people leave the university. Historical data stays intact so you don't lose project records.

2.2.2 Supervisor Profile Management Module

FR-6: Research Interest Publication

Table 2.6: FR-6: Research Interest Publication

Field	Details
Requirement ID	FR-6
Requirement	The system shall allow Supervisors to create and publish their research interests and available project slots.
Source	Project Proposal
Rationale	To help students identify suitable supervisors based on research interests.
Business Rule	Research interests must be published before the submission phase begins.
Dependencies	FR-2
Priority	High

Table 2.6 allows supervisors to publish their research interests so students can find the right match. This has to be done before students start submitting ideas.

FR-7: Supervisor Project Capacity

Table 2.7: FR-7: Supervisor Project Capacity

Field	Details
Requirement ID	FR-7
Requirement	The system shall track active projects per supervisor and enforce the maximum allowed limit.
Source	Project Proposal
Rationale	To ensure that supervisors maintain a manageable workload.
Business Rule	Supervisors cannot exceed the maximum allowed projects (8).
Dependencies	FR-6
Priority	High

Table 2.7 tracks how many projects each supervisor has and prevents them from taking on too many. The limit is 8 active projects maximum.

FR-8: Supervisor Browse and Search

Table 2.8: FR-8: Supervisor Browse and Search

Field	Details
Requirement ID	FR-8
Requirement	The system shall allow Students to browse and search for Supervisors based on research interests and available slots.
Source	Project Proposal
Rationale	To facilitate matching students with appropriate supervisors.
Business Rule	Only active supervisors with available slots appear in search results.

Dependencies FR-6, FR-7

Priority Medium

Table 2.8 lets students search for supervisors based on research areas and availability. Only supervisors with open slots show up in results.

FR-9: Supervisor Project History

Table 2.9: FR-9: Supervisor Project History

Field	Details
Requirement ID	FR-9
Requirement	The system shall maintain a history of past projects supervised by each Supervisor.
Source	Project Proposal
Rationale	To provide insight into a supervisor's experience and project track record.
Business Rule	Only completed projects are included in the history.
Dependencies	FR-6
Priority	Low

Table 2.9 keeps track of each supervisor's past projects so students can see their experience. Only finished projects show up in the history.

2.2.3 Project Idea and Approval Module

FR-10: Project Idea Submission

Table 2.10: FR-10: Project Idea Submission

Field	Details
-------	---------

Requirement ID	FR-10
Requirement	The system shall allow Students to submit project ideas during the designated submission phase.
Source	Project Proposal
Rationale	To capture project proposals in a structured format for evaluation.
Business Rule	Students can submit a maximum of three project ideas.
Dependencies	FR-2, FR-8
Priority	High

Table 2.10 defines how students submit their project ideas during the submission window. They can submit up to 3 ideas maximum.

FR-11: Project Idea Approval Workflow

Table 2.11: FR-11: Project Idea Approval Workflow

Field	Details
Requirement ID	FR-11
Requirement	The system shall implement a workflow for Supervisors to review and respond to submitted project ideas.
Source	Project Proposal
Rationale	To ensure project ideas meet quality standards and supervisor expectations.
Business Rule	Supervisors must provide feedback for rejected or revision-required projects.
Dependencies	FR-10
Priority	High

Table 2.11 sets up the approval process where supervisors review student ideas. They have to give feedback if they reject or ask for revisions.

FR-12: Project Status Tracking

Table 2.12: FR-12: Project Status Tracking

Field	Details
Requirement ID	FR-12
Requirement	The system shall display the current status of project ideas to relevant stakeholders.
Source	Project Proposal
Rationale	To provide transparency in the approval process.
Business Rule	Status updates are displayed in real-time.
Dependencies	FR-10, FR-11
Priority	Medium

Table 2.12 shows project status to everyone involved so they know what's happening. Updates happen in real-time.

FR-13: Project Idea Modification

Table 2.13: FR-13: Project Idea Modification

Field	Details
Requirement ID	FR-13
Requirement	The system shall allow Students to modify project ideas with "Revision Required" status.
Source	Project Proposal

Rationale	To facilitate iterative refinement of project ideas.
Business Rule	Modified ideas must be resubmitted within 7 days of the revision request.
Dependencies	FR-11, FR-12
Priority	Medium

Table 2.13 lets students modify their ideas when supervisors ask for changes. They have 7 days to resubmit after getting revision feedback.

2.2.4 Scope Document Management Module

FR-14: Scope Document Upload

Table 2.14: FR-14: Scope Document Upload

Field	Details
Requirement ID	FR-14
Requirement	The system shall allow Students to upload scope documents for approved project ideas.
Source	Project Proposal
Rationale	To formalize project details through official documentation.
Business Rule	Documents must be in PDF format and not exceed 10MB.
Dependencies	FR-11
Priority	High

Table 2.14 allows students to upload scope documents after their idea gets approved. Files must be PDFs under 10MB.

FR-15: Scope Document Review

Table 2.15: FR-15: Scope Document Review

Field	Details
Requirement ID	FR-15
Requirement	The system shall enable Supervisors to review scope documents and provide feedback or approval.
Source	Project Proposal
Rationale	To ensure documents meet quality standards and project requirements.
Business Rule	Review must be completed within 7 days of submission.
Dependencies	FR-14
Priority	High

Table 2.15 lets supervisors review scope documents and give feedback or approval. They have 7 days to complete the review.

FR-16: Scope Document Version Control

Table 2.16: FR-16: Scope Document Version Control

Field	Details
Requirement ID	FR-16
Requirement	The system shall maintain a version history of scope documents, allowing access to previous versions.
Source	Project Proposal
Rationale	To track document evolution and preserve submission history.
Business Rule	Only the latest approved version is considered official.

Dependencies FR-14, FR-15

Priority Medium

Table 2.16 keeps track of different versions of scope documents. Only the latest approved version is the official one.

FR-17: Document Template Management

Table 2.17: FR-17: Document Template Management

Field	Details
Requirement ID	FR-17
Requirement	The system shall provide downloadable templates for scope documents and other required documentation.
Source	Project Proposal
Rationale	To standardize document formats and ensure completeness.
Business Rule	Template updates do not affect previously submitted documents.
Dependencies	None
Priority	Medium

Table 2.17 provides document templates that students can download to make sure their submissions are in the right format. Template updates don't affect old submissions.

2.2.5 Evaluation Committee Management Module

FR-18: Committee Formation

Table 2.18: FR-18: Committee Formation

Field	Details
-------	---------

Requirement ID	FR-18
Requirement	The system shall allow the Administrator to create evaluation committees with assigned faculty members.
Source	Project Proposal
Rationale	To establish official groups responsible for project evaluation.
Business Rule	Each committee must have at least 2 members.
Dependencies	FR-2
Priority	High

Table 2.18 lets administrators create evaluation committees by assigning faculty members. Each committee needs at least 2 people.

FR-19: Defence Session Scheduling

Table 2.19: FR-19: Defence Session Scheduling

Field	Details
Requirement ID	FR-19
Requirement	The system shall enable scheduling defence sessions with assigned time slots, venues, and committees.
Source	Project Proposal
Rationale	To organize and communicate evaluation schedules efficiently.
Business Rule	Sessions must be scheduled at least 7 days in advance.
Dependencies	FR-18, FR-15
Priority	High

Table 2.19 handles defence session scheduling with committees, venues, and time slots. Sessions need to be scheduled at least 7 days ahead.

FR-20: Evaluation Rubric Management

Table 2.20: FR-20: Evaluation Rubric Management

Field	Details
Requirement ID	FR-20
Requirement	The system shall provide customizable evaluation rubrics for committee members to assess projects.
Source	Project Proposal
Rationale	To standardize project evaluation and ensure fair assessment.
Business Rule	Rubrics must be finalized before the evaluation phase begins.
Dependencies	FR-18
Priority	Medium

Table 2.20 provides evaluation rubrics that committee members use to score projects fairly. Rubrics have to be finalized before evaluations start.

FR-21: Evaluation Feedback Submission

Table 2.21: FR-21: Evaluation Feedback Submission

Field	Details
Requirement ID	FR-21
Requirement	The system shall allow committee members to submit evaluation scores and feedback using the provided rubric.
Source	Project Proposal
Rationale	To capture and store evaluation results for each project.
Business Rule	Feedback must be submitted within 24 hours of the defence session.

Dependencies	FR-19, FR-20
Priority	High

Table 2.21 lets committee members submit their evaluation scores and feedback after defence sessions. They have 24 hours to submit after the defence.

2.2.6 Administrative Dashboard and Reporting Module

FR-22: Administrative Reports

Table 2.22: FR-22: Administrative Reports

Field	Details
Requirement ID	FR-22
Requirement	The system shall generate customizable reports for the Administrator with key project metrics.
Source	Project Proposal
Rationale	To provide insights for administrative decision-making.
Business Rule	Reports can be exported in PDF or Excel format.
Dependencies	Multiple
Priority	Medium

Table 2.22 generates reports for administrators to track project metrics and make decisions. Reports can be exported as PDF or Excel files.

FR-23: Dashboard Visualization

Table 2.23: FR-23: Dashboard Visualization

Field	Details
Requirement ID	FR-23

Requirement	The system shall provide role-specific dashboards displaying relevant metrics and pending tasks.
Source	Project Proposal
Rationale	To provide quick access to important information.
Business Rule	Dashboard content is filtered based on user role.
Dependencies	Multiple
Priority	Medium

Table 2.23 provides custom dashboards for each user type showing relevant info and pending tasks. Content changes based on your role.

2.3 Use Cases

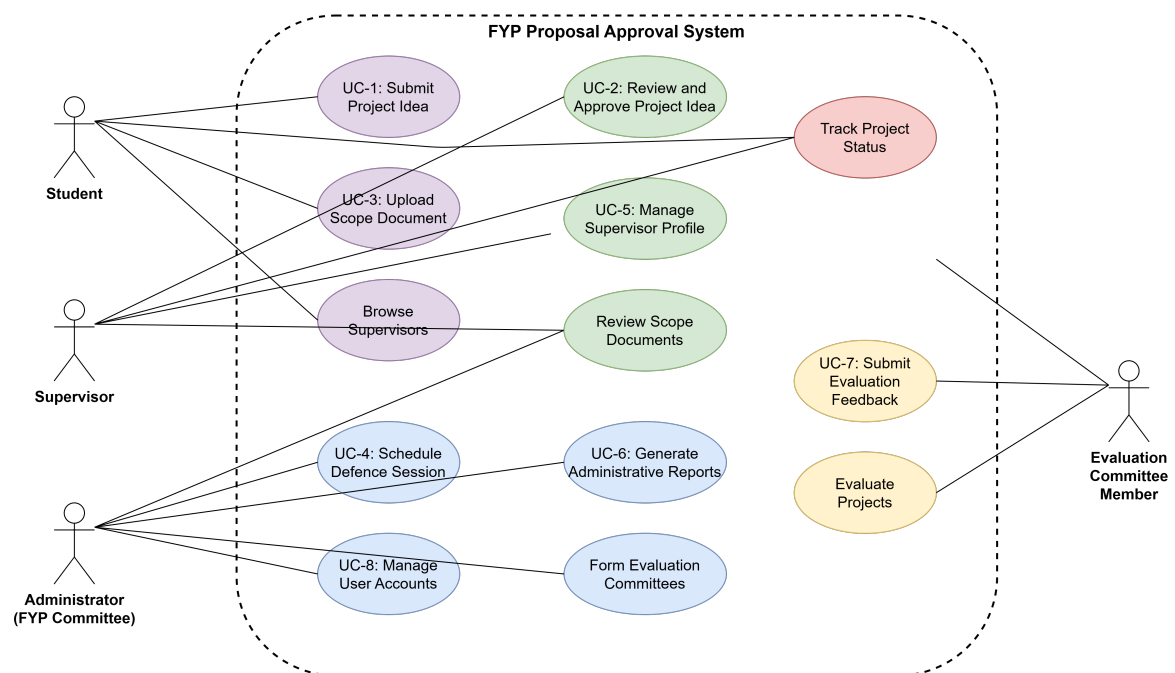


Figure 2.1: System Use Case Diagram

Figure 2.1 shows all the ways different users interact with our FYP system. You can see students, supervisors, admins, and committee members doing their various tasks.

System Actors

Table 2.24: System Actors

Actor		Description
Student		Submits project ideas and scope documents for review and approval.
Supervisor		Reviews project ideas, publishes research interests, and provides feedback.
Administrator (FYP Committee)		Configures system settings, manages users, and schedules defence sessions.
Evaluation Member	Committee	Evaluates projects and submits feedback during defence sessions.

Table 2.24 lists all the different types of users and what they do in the system. These are the main people who will be using our FYP approval system.

Key Use Cases

Table 2.25: Key Use Cases

Use Case ID	Description
UC-1	Submit Project Idea
UC-2	Review and Approve Project Idea
UC-3	Upload Scope Document
UC-4	Schedule Defence Session
UC-5	Submit Evaluation Feedback

Table 2.25 shows the five main workflows in our system. These cover everything from idea submission to final evaluation.

2.3.1 UC-1: Submit Project Idea

Table 2.26: UC-1: Submit Project Idea

Field	Details
Use Case ID	UC-1
Primary Actor	Student
Secondary Actor(s)	Supervisor
Description	A student submits a project idea for supervisor approval during the submission phase.
Trigger	Student initiates the project idea submission process.
Preconditions	• Student is logged into the system. • The idea submission phase is active. • Student has not exceeded the limit of three submissions.
Postconditions	• Project idea stored with status "Pending". • Supervisor can view the submission. • Student can view submission status.
Normal Flow	1. Student navigates to "Submit Project Idea". 2. Fills in project details and selects a supervisor. 3. System validates and stores the submission. 4. System updates project status to "Pending". 5. Confirmation displayed to student.
Alternative Flows	• Save draft for later completion. • Change supervisor before final submission.
Exceptions	• Invalid form data. • Submission phase closed. • Maximum submissions exceeded.

Business Rules	Max 3 ideas; only during submission window; one supervisor per idea; supervisor must have available slots.
Related Requirements	FR-10, FR-6, FR-7

Table 2.26 details how students submit their project ideas to supervisors. This is the main starting point for the whole FYP process.

2.3.2 UC-2: Review and Approve Project Idea

Table 2.27: UC-2: Review and Approve Project Idea

Field	Details
Use Case ID	UC-2
Primary Actor	Supervisor
Secondary Actor(s)	Student
Description	A supervisor reviews a submitted project idea and decides to approve, reject, or request modifications.
Trigger	Supervisor opens a pending project idea for review.
Preconditions	• Supervisor is logged in. • At least one pending idea is assigned. • Review phase is active.
Postconditions	• Status updated (Approved/Rejected/Revision Required). • Student can view decision status. • Approved ideas moved to active list.

Normal Flow

1. Supervisor views pending ideas.
2. Reviews and provides decision and feedback.
3. System updates status.
4. Status visible to student.

Alternative Flows

- Reject idea with feedback.
- Request modification ("Revision Required").
- Defer decision to later.

Exceptions

- Review period expired.
- Supervisor capacity exceeded.
- Student withdrew submission.

Business Rules Max 8 active projects; feedback required; review within 7 days.

Related Requirements FR-11, FR-7, FR-12

Table 2.27 shows how supervisors review student ideas and decide whether to approve or reject them. This is where quality control happens.

2.3.3 UC-3: Upload Scope Document

Table 2.28: UC-3: Upload Scope Document

Field	Details
Use Case ID	UC-3
Primary Actor	Student
Secondary Actor(s)	Supervisor
Description	A student uploads a scope document for an approved project idea.

Trigger	Student initiates scope document upload.
Preconditions	• Student logged in. • Approved project idea exists. • Scope submission phase active.
Postconditions	• Document stored with "Pending Review". • Supervisor can view the document. • Version record created.
Normal Flow	1. Student selects approved project. 2. Uploads PDF file. 3. System validates and stores document. 4. Document available for supervisor review.
Alternative Flows	• Replace existing version. • Download official template before upload.
Exceptions	• Invalid file format. • File exceeds 10MB.
Business Rules	PDF only, 10MB, latest approved version considered official.
Related Requirements	Re- FR-14, FR-15, FR-16

Table 2.28 covers how students upload their scope documents after idea approval. This moves projects to the formal documentation stage.

2.3.4 UC-4: Schedule Defence Session

Table 2.29: UC-4: Schedule Defence Session

Field	Details
-------	---------

Use Case ID	UC-4
Primary Actor	Administrator (FYP Committee)
Secondary Actor(s)	Supervisor, Student, Evaluation Committee Members
Description	The administrator schedules defence sessions for projects with approved scope documents.
Trigger	Administrator initiates defence scheduling.
Preconditions	• Administrator logged in. • Approved scope documents available. • Evaluation committees formed.
Postconditions	• Defence session stored. • All participants can view schedule. • Project marked as "Defence Scheduled".
Normal Flow	1. Admin opens "Defence Scheduling". 2. Selects projects and assigns committees. 3. Confirms schedule and venue. 4. System stores schedule.
Alternative Flows	• Batch scheduling. • Reschedule defence.
Exceptions	• No available committee. • Venue conflict. • Insufficient notice (≥7 days).
Business Rules	Minimum 7 days advance; 2 committee members; 3 sessions per venue/day.

Related Requirements FR-19, FR-18, FR-23

Table 2.29 explains how administrators schedule defence sessions with committees and venues. The conflict checking is automated to prevent double-booking.

2.3.5 UC-5: Submit Evaluation Feedback

Table 2.30: UC-5: Submit Evaluation Feedback

Field	Details
Use Case ID	UC-5
Primary Actor	Evaluation Committee Member
Secondary Actor(s)	Student, Supervisor
Description	Committee members submit scores and feedback after defence using rubrics.
Trigger	Member opens evaluation form after defence.
Preconditions	• Member logged in. • Defence session scheduled. • Rubric finalised.
Postconditions	• Scores and feedback stored. • Results available to stakeholders. • Final result calculated.
Normal Flow	1. Member opens "My Evaluations". 2. Selects completed session. 3. Fills rubric and submits. 4. System records evaluation.

Alternative	
Flows	• Save as draft. • Review project details before scoring.
Exceptions	• Incomplete evaluation. • Submission deadline missed.
Business Rules	Feedback due within 24 hours; all criteria must be scored; written comments required.
Related Re-quirements	FR-21, FR-20

Table 2.30 shows how committee members submit their evaluation scores after defence sessions. This completes the whole FYP approval cycle.

2.4 Non-Functional Requirements

NFR-1: Dashboard Load Time

Table 2.31: NFR-1: Dashboard Load Time

Field	Details
Requirement ID	NFR-1
Category	Performance
Requirement	The system shall load the main dashboard within 3 seconds under standard network conditions.
Rationale	To ensure a responsive and user-friendly experience.
Priority	High

Table 2.31 sets the dashboard load time at 3 seconds max. This keeps the system responsive for users.

NFR-2: Concurrent Users Handling

Table 2.32: NFR-2: Concurrent Users Handling

Field	Details
Requirement ID	NFR-2
Category	Performance
Requirement	The system shall handle at least 200 concurrent users without performance degradation.
Rationale	To ensure system scalability during high load such as submission deadlines.
Priority	High

Table 2.32 ensures the system can handle 200 users at the same time. This is important during submission deadlines when everyone's online.

NFR-3: Query Response Time

Table 2.33: NFR-3: Query Response Time

Field	Details
Requirement ID	NFR-3
Category	Performance
Requirement	Database queries shall execute within one second for normal operations.
Rationale	To provide quick feedback and efficient data processing.
Priority	High

Table 2.33 requires database queries to finish within 1 second. This keeps everything running smoothly.

NFR-4: Horizontal Scalability

Table 2.34: NFR-4: Horizontal Scalability

Field	Details
Requirement ID	NFR-4
Category	Scalability
Requirement	The system architecture shall support horizontal scaling to handle increased future load.
Rationale	To allow system growth without major redesign.
Priority	Medium

Table 2.34 makes sure the system can grow by adding more servers. This is good for future expansion.

NFR-5: Schema Flexibility

Table 2.35: NFR-5: Schema Flexibility

Field	Details
Requirement ID	NFR-5
Category	Scalability
Requirement	The database schema shall allow addition of new entities and relationships without breaking existing functionality.
Rationale	To support evolving academic and administrative requirements.
Priority	Medium

Table 2.35 allows the database to be modified without breaking existing features. This helps when requirements change.

NFR-6: System Availability

Table 2.36: NFR-6: System Availability

Field	Details
Requirement ID	NFR-6
Category	Availability
Requirement	The system shall maintain at least 99.5% uptime excluding scheduled maintenance.
Rationale	To ensure reliable access during key academic periods.
Priority	High

Table 2.36 requires 99.5

NFR-7: Retry on Network Failure

Table 2.37: NFR-7: Retry on Network Failure

Field	Details
Requirement ID	NFR-7
Category	Reliability
Requirement	The system shall automatically retry any submission or upload that fails due to temporary network issues.
Rationale	To prevent data loss and ensure successful submissions.
Priority	Medium

Table 2.37 automatically retries failed uploads due to network problems. This prevents losing student work.

NFR-8: File Integrity Verification

Table 2.38: NFR-8: File Integrity Verification

Field	Details
Requirement ID	NFR-8
Category	Reliability
Requirement	All uploaded files shall be checksum-verified to ensure integrity during transfer.
Rationale	To guarantee uploaded documents are complete and uncorrupted.
Priority	Medium

Table 2.38 checks that uploaded files aren't corrupted during upload. This ensures document integrity.

NFR-9: Secure Password Storage

Table 2.39: NFR-9: Secure Password Storage

Field	Details
Requirement ID	NFR-9
Category	Security
Requirement	All user credentials shall be encrypted and stored using bcrypt hashing with unique salt per user.
Rationale	To protect sensitive user credentials against unauthorized access.
Priority	High

Table 2.39 uses bcrypt encryption to store passwords securely. This protects user accounts from hackers.

NFR-10: Encrypted Communication

Table 2.40: NFR-10: Encrypted Communication

Field	Details
Requirement ID	NFR-10
Category	Security
Requirement	The system shall use HTTPS with TLS 1.2 or higher for all client-server communications.
Rationale	To prevent data interception or man-in-the-middle attacks.
Priority	High

Table 2.40 requires HTTPS encryption for all communications. This protects data transmission from eavesdropping.

NFR-11: Input Validation and Protection

Table 2.41: NFR-11: Input Validation and Protection

Field	Details
Requirement ID	NFR-11
Category	Security
Requirement	The system shall validate all inputs to prevent SQL Injection, XSS, and CSRF attacks.
Rationale	To ensure data integrity and application security.
Priority	High

Table 2.41 protects against common web attacks like SQL injection and XSS. This keeps the system secure from malicious input.

NFR-12: Session Timeout

Table 2.42: NFR-12: Session Timeout

Field	Details
Requirement ID	NFR-12
Category	Security
Requirement	User sessions shall automatically expire after 30 minutes of inactivity.
Rationale	To prevent unauthorized access from unattended systems.
Priority	High

Table 2.42 automatically logs users out after 30 minutes of inactivity. This prevents unauthorized access from unattended computers.

NFR-13: Password Reset Expiry

Table 2.43: NFR-13: Password Reset Expiry

Field	Details
Requirement ID	NFR-13
Category	Security
Requirement	Password reset links shall expire within 15 minutes of generation.
Rationale	To minimize risk of compromised reset links.
Priority	Medium

Table 2.43 makes password reset links expire after 15 minutes. This reduces security risks from old reset emails.

NFR-14: Role-Based Access Control

Table 2.44: NFR-14: Role-Based Access Control

Field	Details
Requirement ID	NFR-14
Category	Security
Requirement	Access to each module shall be restricted based on the user's role in the system.
Rationale	To enforce proper authorization boundaries.
Priority	High

Table 2.44 restricts access based on user roles to prevent unauthorized actions. Students can't access admin functions, etc.

NFR-15: Field-Level Validation

Table 2.45: NFR-15: Field-Level Validation

Field	Details
Requirement ID	NFR-15
Category	Usability
Requirement	The system shall display validation and error messages at the field level for all forms.
Rationale	To enhance user experience and reduce input errors.
Priority	Medium

Table 2.45 shows error messages right next to form fields where problems occur. This makes it easier for users to fix mistakes.

NFR-16: Consistent UI Design

Table 2.46: NFR-16: Consistent UI Design

Field	Details
Requirement ID	NFR-16
Category	Usability
Requirement	The system shall maintain consistent navigation, color scheme, and component layout throughout.
Rationale	To ensure intuitive and professional user interface.
Priority	Medium

Table 2.46 keeps the same design style throughout the whole system. This makes navigation intuitive and professional-looking.

NFR-17: Mobile Responsiveness

Table 2.47: NFR-17: Mobile Responsiveness

Field	Details
Requirement ID	NFR-17
Category	Usability
Requirement	The web interface shall adapt to different screen sizes with responsive layouts.
Rationale	To ensure accessibility across mobile, tablet, and desktop devices.
Priority	Medium

Table 2.47 makes sure the system works properly on phones, tablets, and computers. The layout adjusts automatically to different screen sizes.

NFR-18: Accessibility Compliance

Table 2.48: NFR-18: Accessibility Compliance

Field	Details
Requirement ID	NFR-18
Category	Usability
Requirement	The user interface shall comply with WCAG 2.1 AA accessibility guidelines.
Rationale	To support users with visual and physical impairments.
Priority	Low

Table 2.48 follows accessibility standards to make the system usable for people with disabilities. This ensures everyone can use the system.

NFR-19: MVC Architecture

Table 2.49: NFR-19: MVC Architecture

Field	Details
Requirement ID	NFR-19
Category	Maintainability
Requirement	The application shall follow MVC architecture using Laravel 11 framework.
Rationale	To enhance maintainability and modularity of the system.
Priority	High

Table 2.49 uses the MVC pattern with Laravel framework for better code organization. This makes the system easier to maintain and update.

NFR-20: Configuration Management

Table 2.50: NFR-20: Configuration Management

Field	Details
Requirement ID	NFR-20
Category	Maintainability
Requirement	All environment-specific configurations shall be stored in the .env file.
Rationale	To separate code from configuration and simplify deployment.
Priority	High

Table 2.50 keeps all configuration settings in separate .env files. This makes deployment to different environments much easier.

NFR-21: Documentation Updates

Table 2.51: NFR-21: Documentation Updates

Field	Details
Requirement ID	NFR-21
Category	Maintainability
Requirement	The technical and user documentation shall be updated with every major release.
Rationale	To ensure documentation accuracy and developer onboarding.
Priority	Medium

Table 2.51 requires documentation updates with each major release. This keeps everything current for developers and users.

NFR-22: OS Portability

Table 2.52: NFR-22: OS Portability

Field	Details
Requirement ID	NFR-22
Category	Portability
Requirement	The system shall operate seamlessly on Linux and Windows environments.
Rationale	To support university server infrastructure flexibility.
Priority	Medium

Table [2.52](#) makes sure the system works on both Linux and Windows servers. This gives universities deployment flexibility.

NFR-23: Browser Compatibility

Table 2.53: NFR-23: Browser Compatibility

Field	Details
Requirement ID	NFR-23
Category	Portability
Requirement	The system shall support Chrome, Edge, and Firefox (latest two versions).
Rationale	To ensure consistent experience across popular browsers.
Priority	Medium

Table [2.53](#) supports the most popular browsers to ensure wide compatibility. Users won't have issues regardless of their browser choice.

NFR-24: Automated Backups

Table 2.54: NFR-24: Automated Backups

Field	Details
Requirement ID	NFR-24
Category	Backup and Recovery
Requirement	Daily automated database backups shall be performed at 2:00 AM server time.
Rationale	To minimize data loss in case of system failure.
Priority	High

Table 2.54 backs up the database every night at 2 AM automatically. This protects against data loss if something goes wrong.

NFR-25: Recovery Time Objective

Table 2.55: NFR-25: Recovery Time Objective

Field	Details
Requirement ID	NFR-25
Category	Backup and Recovery
Requirement	System recovery shall be completed within 30 minutes of initiating restore procedures.
Rationale	To ensure minimal downtime after incidents.
Priority	Medium

Table 2.55 sets the recovery time to 30 minutes maximum. This minimizes downtime when restoring from backups.

NFR-26: Activity Logging

Table 2.56: NFR-26: Activity Logging

Field	Details
Requirement ID	NFR-26
Category	Audit and Logging
Requirement	Every critical user action shall be logged with timestamp and user ID.
Rationale	To maintain audit trails and support accountability.
Priority	High

Table [2.56](#) logs all important user actions for audit purposes. This helps track what happened and who did it.

NFR-27: Log Retention Policy

Table 2.57: NFR-27: Log Retention Policy

Field	Details
Requirement ID	NFR-27
Category	Audit and Logging
Requirement	System logs shall be retained for one year and secured from modification.
Rationale	To comply with audit standards and ensure traceability.
Priority	Medium

Table [2.57](#) keeps system logs for one year and protects them from tampering. This supports compliance and investigation needs.

NFR-28: Standards Compliance

Table 2.58: NFR-28: Standards Compliance

Field	Details
Requirement ID	NFR-28
Category	Compliance
Requirement	The system shall comply with ISO/IEC 27001 and university IT security policies.
Rationale	To meet institutional and international compliance standards.
Priority	High

Table [2.58](#) meets international security standards and university IT policies. This ensures the system follows proper security practices and compliance requirements.

3 Design

3.1 Data Flow Diagrams (DFDs)

3.1.1 Zero Level DFD (Context Diagram)

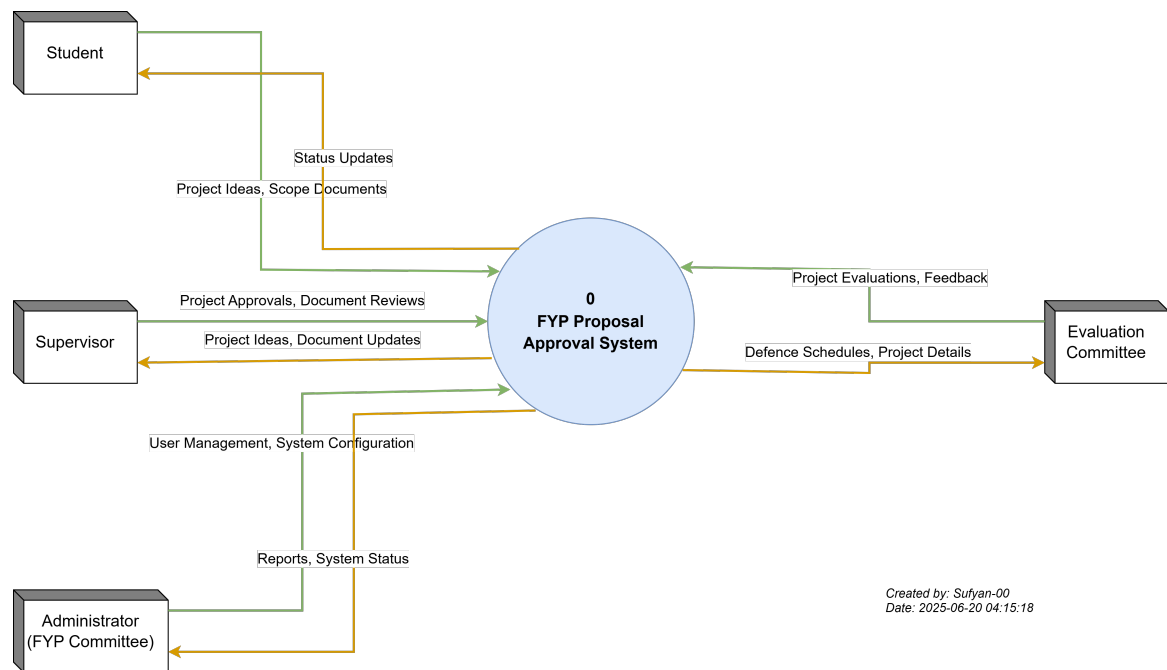


Figure 3.1: DFD Level 0

Figure 3.1 shows the big picture of our FYP system and how different people interact with it. You can see students, supervisors, admins, and committee members all connecting to the main system.

3.1.2 First Level DFDs

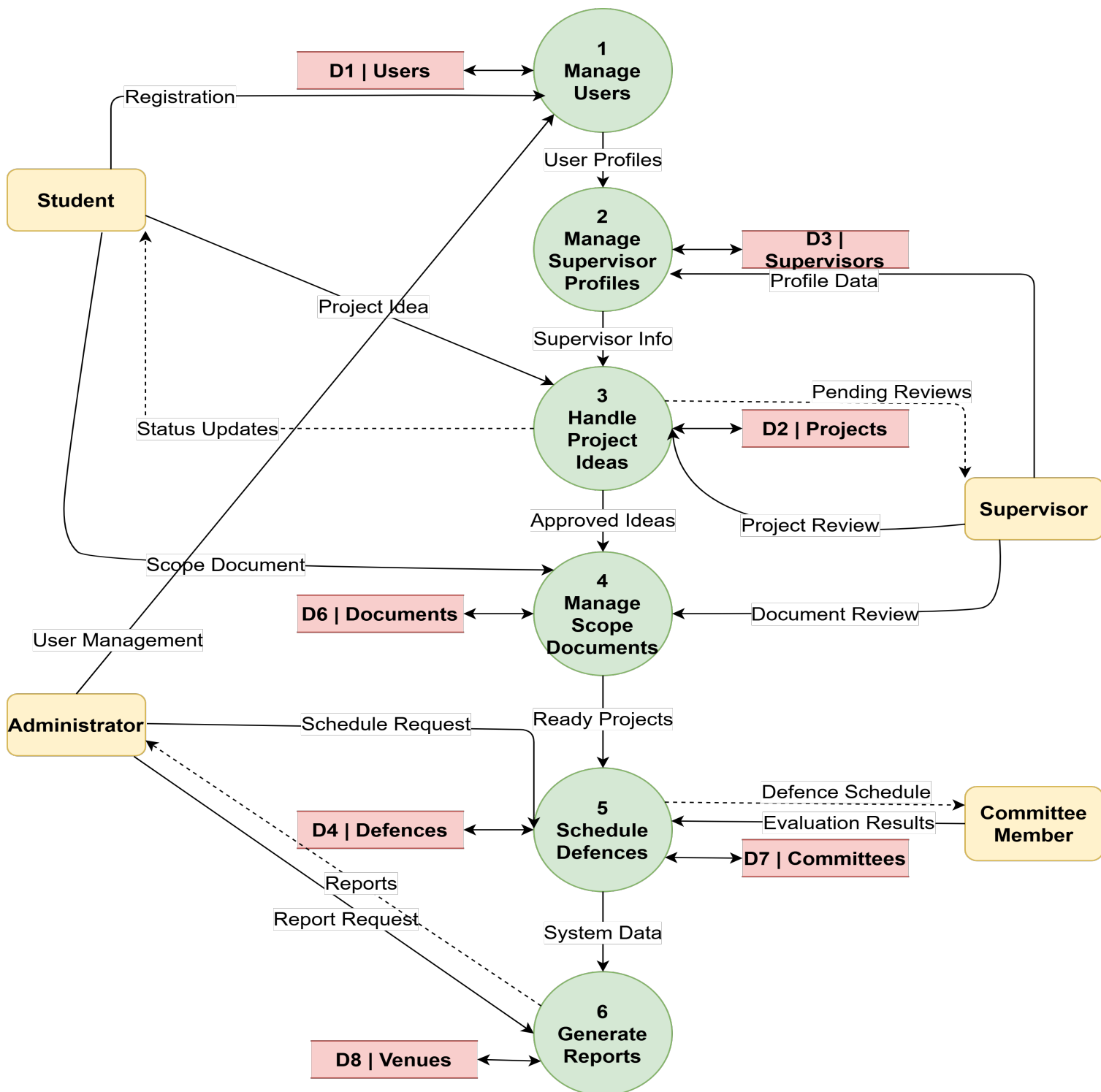


Figure 3.2: DFD Level 1

Figure 3.2 breaks down the system into its main parts like user management, project submission, and evaluation processes. This gives you a better idea of what happens

inside the system.

3.1.3 Second Level DFDs

Project Idea Workflow

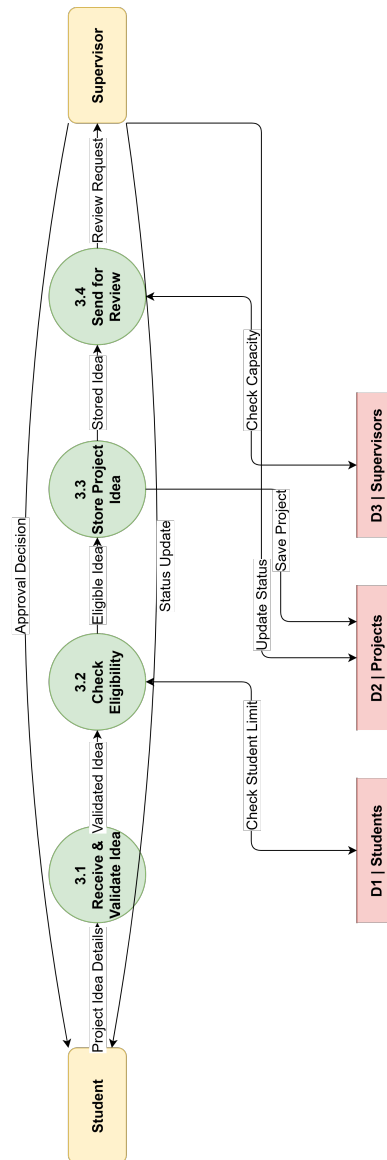


Figure 3.3: DFD Level 2 – Project idea submission

Figure 3.3 shows exactly what happens when a student submits a project idea step by step. You can see all the validation and storage processes that keep everything organized.

Defence Scheduling

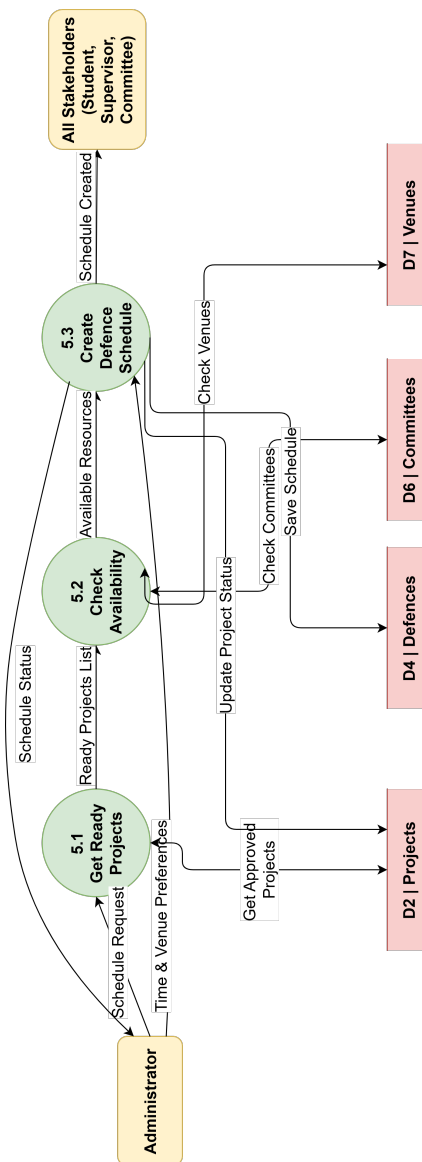


Figure 3.4: DFD Level 2 – Defence Scheduling

Figure 3.4 details how defence sessions get scheduled with all the committee and venue coordination. This used to be a nightmare to do manually but now the system handles the complexity.

3.2 Interaction Diagram

3.2.1 Sequence Diagrams

Submit Project Idea

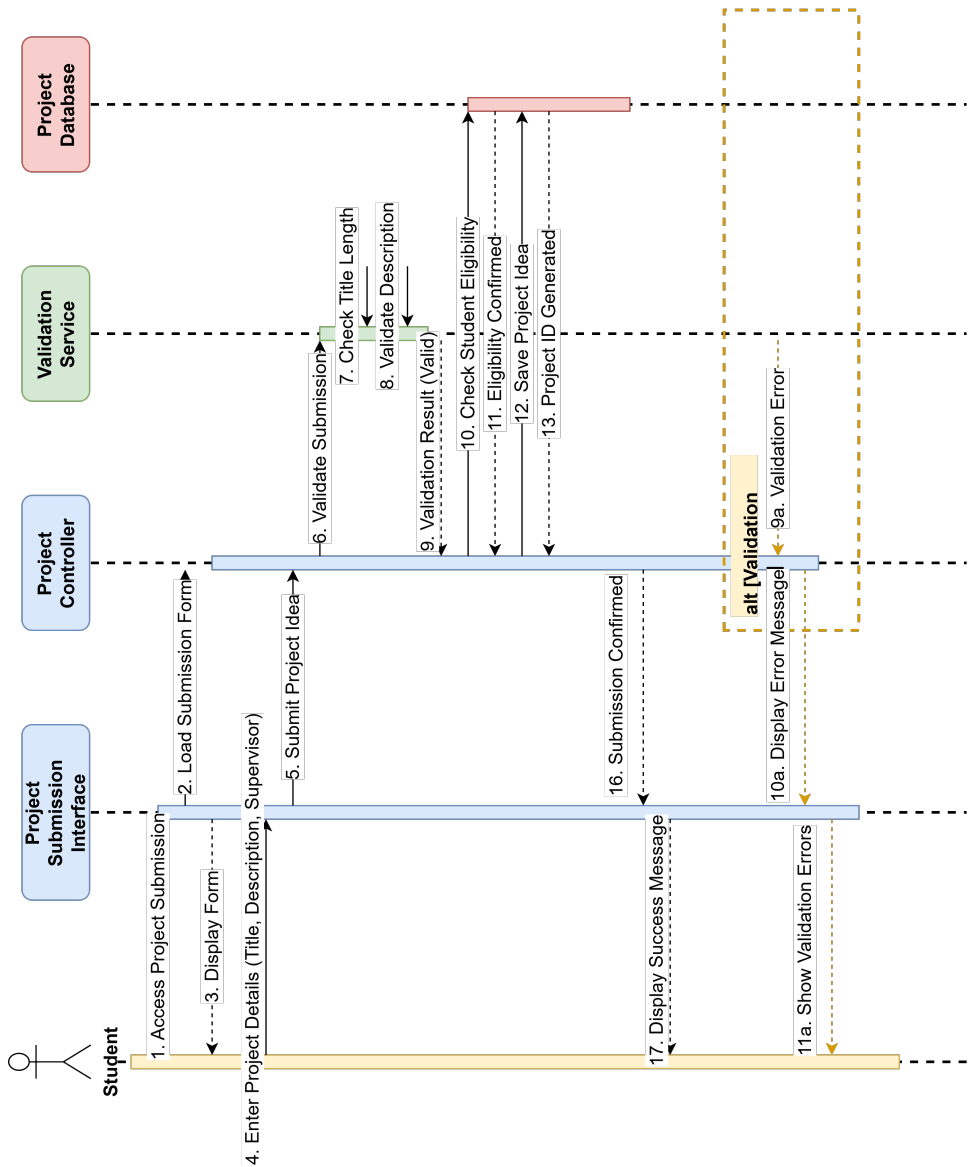


Figure 3.5: Sequence Diagram – Idea submission

Figure 3.5 shows the timeline of what happens when a student submits their project idea. You can see how the system validates everything and stores it properly.

Approve Project Idea

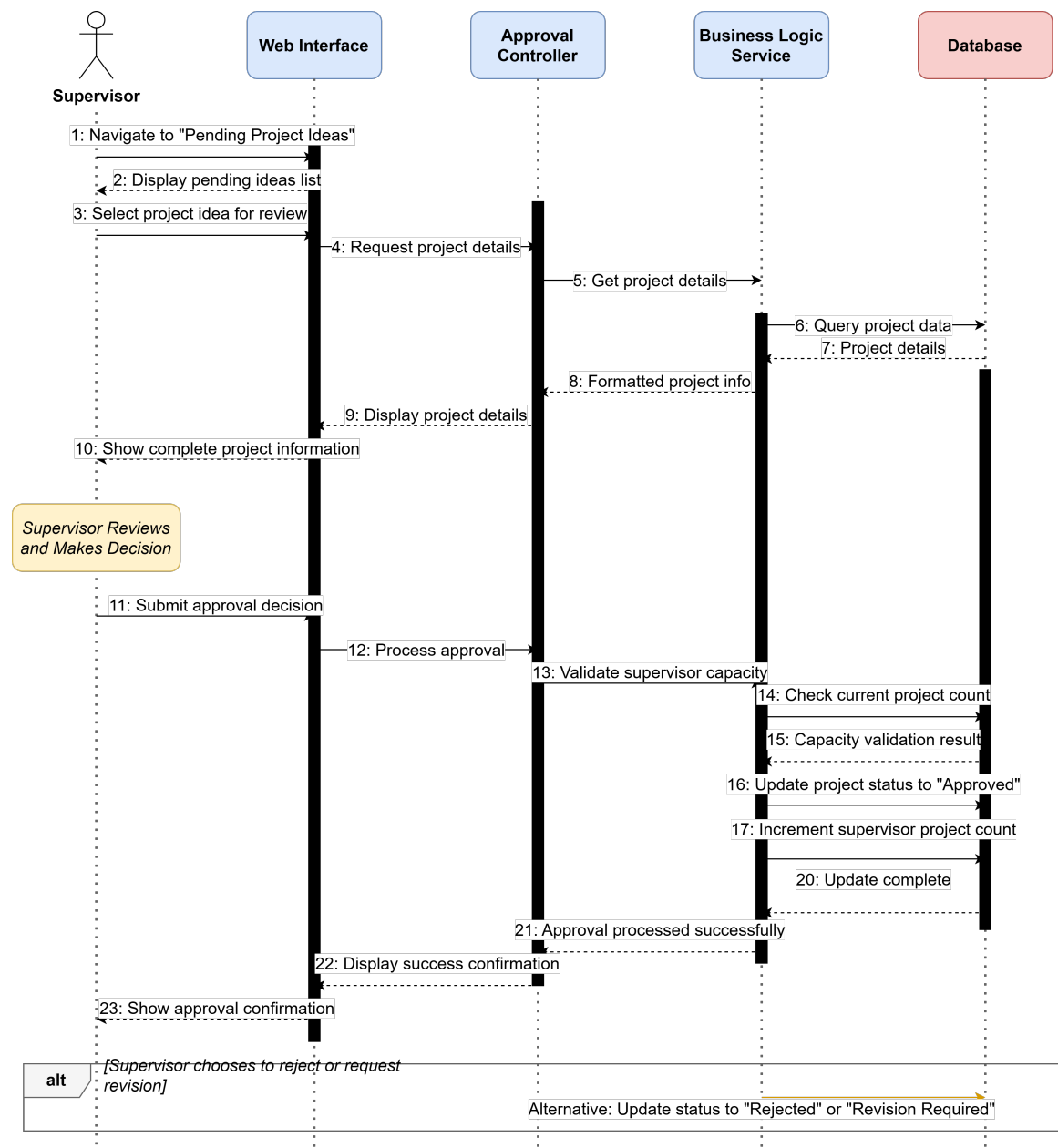


Figure 3.6: Sequence Diagram – Idea approval

Figure 3.6 demonstrates how supervisors review and approve project ideas in the system. It also shows how capacity tracking gets updated automatically.

Schedule Defence

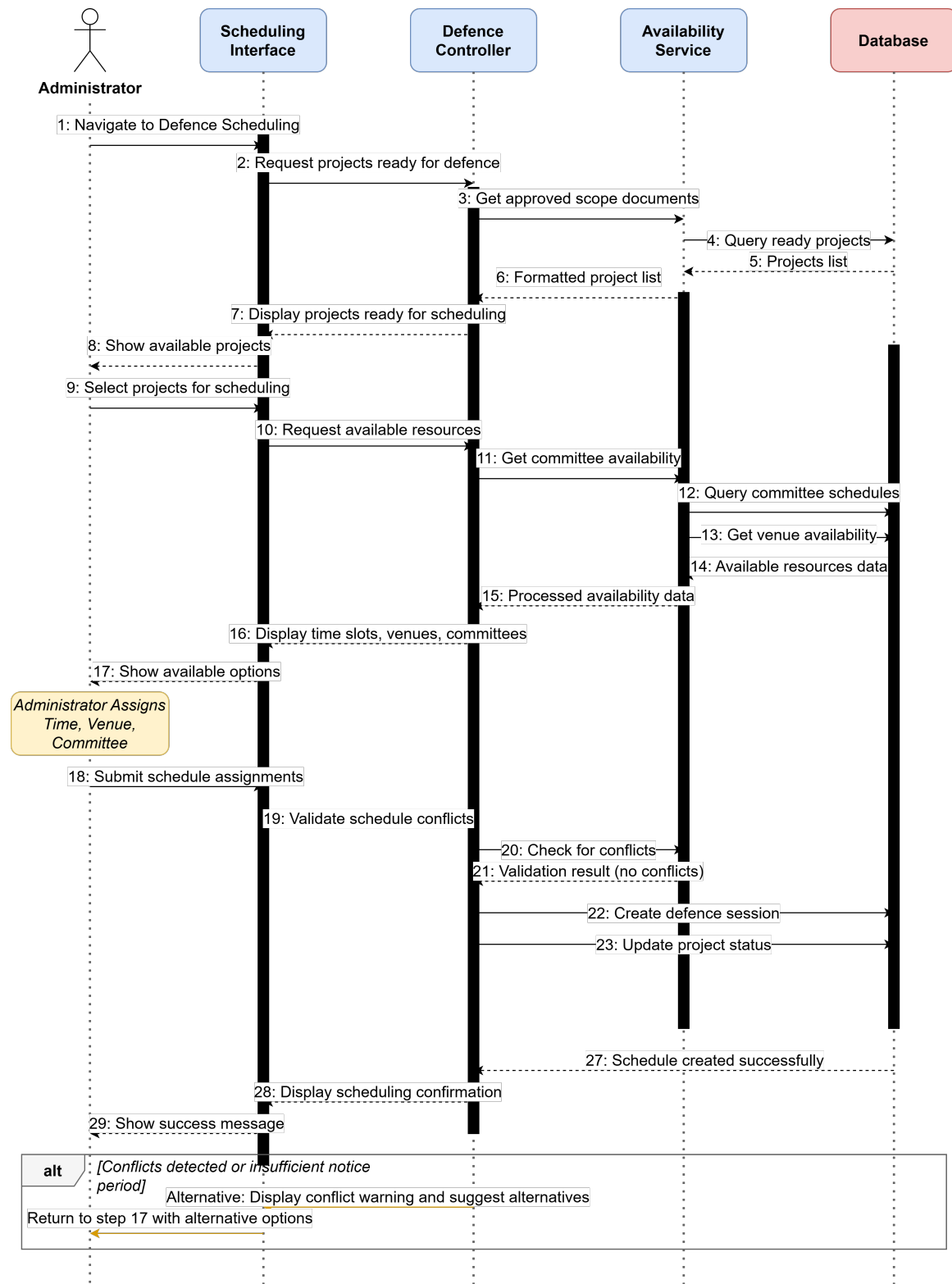


Figure 3.7: Sequence Diagram – Scheduling defence

Figure 3.7 illustrates the complex process of scheduling defence sessions with conflict checking. The system coordinates multiple people and resources automatically.

3.2.2 Collaboration Diagrams

User authentication flow

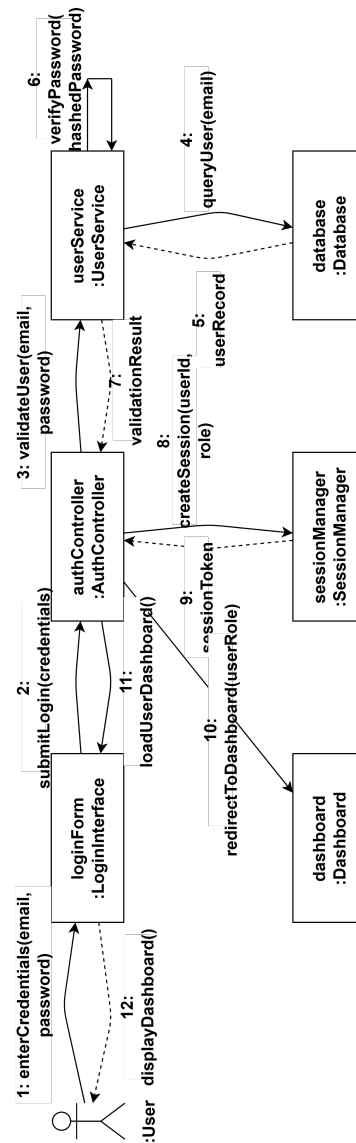


Figure 3.8: Collaboration Diagram – User auth

Figure 3.8 shows how different parts of the system work together when a user logs in. The authentication controller and session management handle all the security stuff.

Document upload and validation

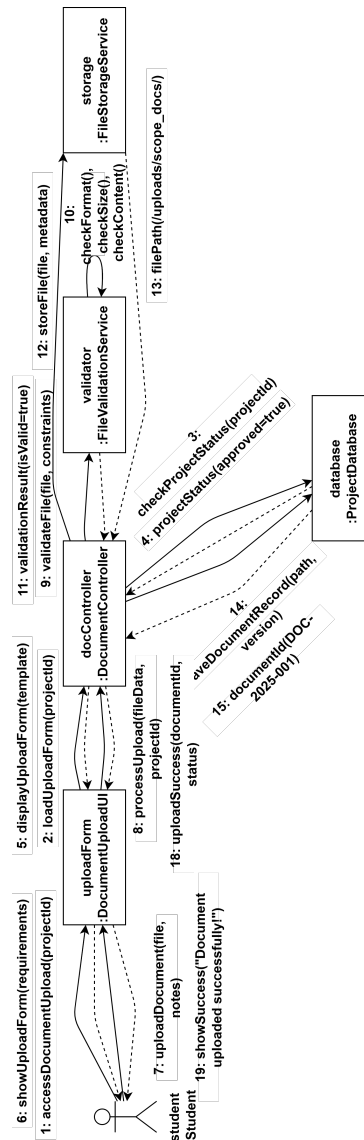


Figure 3.9: Collaboration Diagram – Document upload and validation

Figure 3.9 demonstrates how the document upload system works with validation and file storage. Multiple components work together to make sure files are handled properly.

Evaluation feedback submission

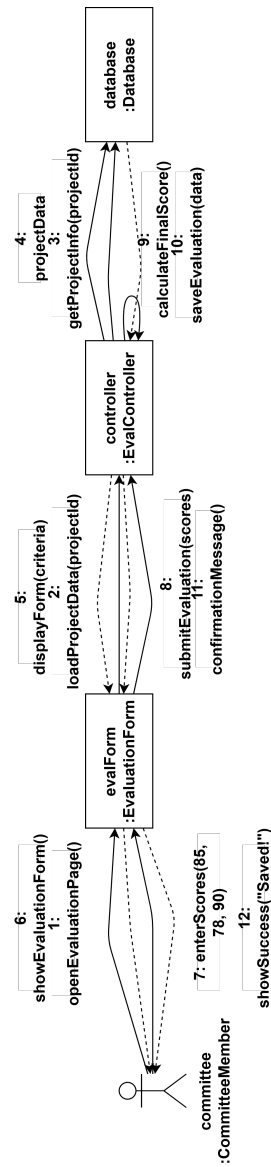


Figure 3.10: Collaboration Diagram – Evaluation Feedback

Figure 3.10 shows how committee members submit their project evaluations and scores. The system handles rubric management and calculates final results automatically.

4 Testing

4.1 Test Cases

TC-001 – User Login

Table 4.1: TC-001 – User Login

Field	Details
Title	Verify user login with valid credentials
Module	User Management
Priority	High
Preconditions	1) User account exists in database 2) User is on login page
Test Steps	1) Enter valid email 2) Enter correct password 3) Click "Login"
Expected Result	1) User authenticated 2) Redirect to role-specific dashboard 3) Session created
Actual Result	Authenticated; redirected to student dashboard; session created with 30-minute timeout
Status	Pass
Tested By	Sufyan-00
Date	2025-08-20

Table 4.1 shows the basic login test to make sure users can get into the system with their email and password. This is pretty important since nobody can use the system without logging in first.

TC-002 – Submit Project Idea (Valid)

Table 4.2: TC-002 – Submit Project Idea (Valid)

Field	Details
Title	Verify project idea submission with valid data
Module	Project Idea and Approval
Priority	High
Preconditions	Student logged in; submission phase active; student has ≥3 active submissions
Test Steps	Navigate to "Submit Project Idea"; fill required fields; select available supervisor; submit
Expected Result	Validation passes; idea saved "Pending"; status visible in dashboard
Actual Result	As expected; status visible
Status	Pass
Tested By	Sufyan-00
Date	2025-08-21

Table 4.2 tests the main functionality where students submit their project ideas. This is the core feature of our whole system.

TC-003 – Approve Project Idea

Table 4.3: TC-003 – Approve Project Idea

Field	Details
Title	Verify supervisor can approve project idea
Module	Project Idea and Approval
Priority	High

Preconditions	Supervisor logged in; pending idea exists; supervisor has ≥8 active projects
Test Steps	Navigate to pending ideas; review; approve; confirm
Expected Result	Status "Approved"; supervisor count increments; scope upload enabled
Actual Result	As expected; count increased 2 to 3; upload enabled
Status	Pass
Tested By	Sufyan-00
Date	2025-08-22

Table 4.3 checks if supervisors can approve student projects and if the system updates everything properly. It also makes sure the capacity tracking works correctly.

TC-004 – Schedule Defence Session

Table 4.4: TC-004 – Schedule Defence Session

Field	Details
Title	Verify administrator can schedule defence session
Module	Evaluation Committee Management
Priority	High
Preconditions	Admin logged in; project has approved scope; committee exists; venue/time available
Test Steps	Defence Scheduling; select project; set datetime/venue; assign committee; schedule
Expected Result	Session created; calendar updated; status "Defence Scheduled"; no conflicts
Actual Result	As expected; no conflicts
Status	Pass
Tested By	Sufyan-00

Date	2025-08-23
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Table 4.4 tests the defence scheduling feature where administrators coordinate sessions with committees and venues. The conflict detection is really important here.

TC-005 – Upload Scope Document

Table 4.5: TC-005 – Upload Scope Document

Field	Details
Title	Verify student can upload scope document
Module	Scope Document Management
Priority	High
Preconditions	Student logged in; idea approved; valid PDF ;10MB
Test Steps	Approved project; upload scope; select PDF; add notes; upload
Expected Result	Validation passes; uploaded; version history; status "Pending Review"
Actual Result	PDF 2. 3MB uploaded; history v1.0; status updated
Status	Pass
Tested By	Sufyan-00
Date	2025-08-23

Table 4.5 makes sure students can upload their scope documents after project approval. The version tracking is also tested here.

TC-006 – File Size Validation

Table 4.6: TC-006 – File Size Validation

Field	Details
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Title	Verify system rejects files exceeding size limit
Module	Scope Document Management
Priority	Medium
Preconditions	Student logged in; attempts upload >10MB
Test Steps	Select large file; upload
Expected Result	Error; reject; no file stored
Actual Result	Error: 12. 5MB exceeds 10MB; blocked; no save
Status	Pass
Tested By	Sufyan-00
Date	2025-08-23

Table 4.6 tests that our system properly rejects files that are too big. This prevents storage issues and keeps uploads reasonable.

TC-007 – Session Timeout

Table 4.7: TC-007 – Session Timeout

Field	Details
Title	Verify session timeout after 30 minutes inactivity
Module	User Management (Security)
Priority	High
Preconditions	User logged in
Test Steps	Idle 30 minutes; perform action
Expected Result	Session expires; redirect login; message shown
Actual Result	As expected; flash message shown
Status	Pass

Tested By	Sufyan-00
Date	2025-08-24

Table 4.7 checks that user sessions automatically log out after 30 minutes of inactivity. This is a basic security feature.

TC-008 – SQL Injection Protection

Table 4.8: TC-008 – SQL Injection Protection

Field	Details
Title	Verify system protects against SQL injection
Module	Security
Priority	Critical
Preconditions	Access to any DB form
Test Steps	Enter SQL payload; submit; observe
Expected Result	Sanitized; no SQL executed; app secure
Actual Result	Inputs sanitized; payloads literal; no compromise
Status	Pass
Tested By	Sufyan-00
Date	2025-08-25

Table 4.8 tests protection against SQL injection attacks by trying malicious database queries. This is critical security stuff.

TC-009 – Project Capacity Management

Table 4.9: TC-009 – Project Capacity Management

Field	Details
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Title	Verify supervisor project capacity tracking and enforcement
Module	Supervisor Profile Management
Priority	High
Preconditions	Supervisor logged in; has active projects; max capacity set
Test Steps	View My Projects; check counts and available slots; try to accept at capacity
Expected Result	Correct counts; warnings; cannot exceed max
Actual Result	3/6 active; 3 slots remaining; disabled at max; 50% utilization bar
Status	Pass
Tested By	Sufyan-00
Date	2025-08-26

Table 4.9 makes sure the system properly tracks how many projects each supervisor has. It prevents them from taking on too many projects.

TC-010 – Supervisor Profile Creation

Table 4.10: TC-010 – Supervisor Profile Creation

Field	Details
Title	Verify supervisor can create and update profile with research interests
Module	Supervisor Profile Management
Priority	High
Preconditions	Supervisor logged in; profile absent/incomplete
Test Steps	Edit Profile; fill research interests; set capacity (1–8); office; specialization; save

Expected Result	Form loads; inputs valid; saved; visible to students
Actual Result	Saved; capacity 6; profile visible
Status	Pass
Tested By	Sufyan-00
Date	2025-08-26

Table 4.10 tests that supervisors can create their profiles with research interests and capacity settings. Students need this info to pick the right supervisor.

TC-011 – View Project History

Table 4.11: TC-011 – View Project History

Field	Details
Title	Verify supervisor can view complete project supervision history
Module	Supervisor Profile Management
Priority	Medium
Preconditions	Supervisor logged in; completed projects exist
Test Steps	View History; filter by year; view details
Expected Result	Grouped Active/Completed; filters work; details show students/outcomes
Actual Result	8 total; 3 Active, 5 Completed; 2024–25 shows 3; details show grades
Status	Pass
Tested By	Sufyan-00
Date	2025-08-26

Table 4.11 checks if supervisors can see their complete project history with filtering options. This helps with transparency and track records.

4.2 Test Reports

4.2.1 Test Execution Summary

Table 4.12: Execution Summary

Metric	Count	Percentage
Total Test Cases	11	100%
Test Cases Executed	11	100%
Test Cases Passed	11	100%
Test Cases Failed	0	0%
Test Cases Blocked	0	0%
Test Cases Not Executed	0	0%

Table 4.12 shows our overall testing results - we ran all 11 test cases and they all passed. Pretty good success rate!

4.2.2 Defect Summary

Table 4.13: Defect Summary

Severity	Open	Closed	Total
Critical	0	0	0
High	0	0	0
Medium	0	2	2
Low	1	4	5
Total	1	6	7

Table 4.13 tracks all the bugs we found during testing. Most are fixed, with just one minor issue still open.

4.2.3 Module-wise Test Coverage

Table 4.14: Module-wise Test Coverage

Module	Total Tests	Passed	Failed	Pass Rate
User Management	2	2	0	100%
Supervisor Profile Management	3	3	0	100%
Project Idea and Approval	2	2	0	100%
Scope Document Management	2	2	0	100%
Evaluation Committee Management	1	1	0	100%
Security	1	1	0	100%

Table 4.14 breaks down testing by different parts of the system. Every module got 100

4.2.4 Risk Assessment

Table 4.15: Risk Assessment

Risk Level	Area	Impact	Mitigation Status
Low	Core Functionality	System stable	Complete
Low	Data Security	Basic protection verified	Complete
Medium	Performance at Scale	User experience	Planned

Table 4.15 shows the main risks we identified and how we're handling them. The only concern left is performance when lots of users are online.

5 Implementation

5.1 Database Design

5.1.1 Overview

The system uses a normalized relational schema (MySQL 8.0) following an MVC data access pattern. Core entities include Users, Roles, Supervisors, Projects (ideas), Scope Documents, Committees, Defence Sessions, and Evaluations. All timestamps are stored in UTC; soft deletes are enabled where noted.

5.1.2 Entity-Relationship Diagram

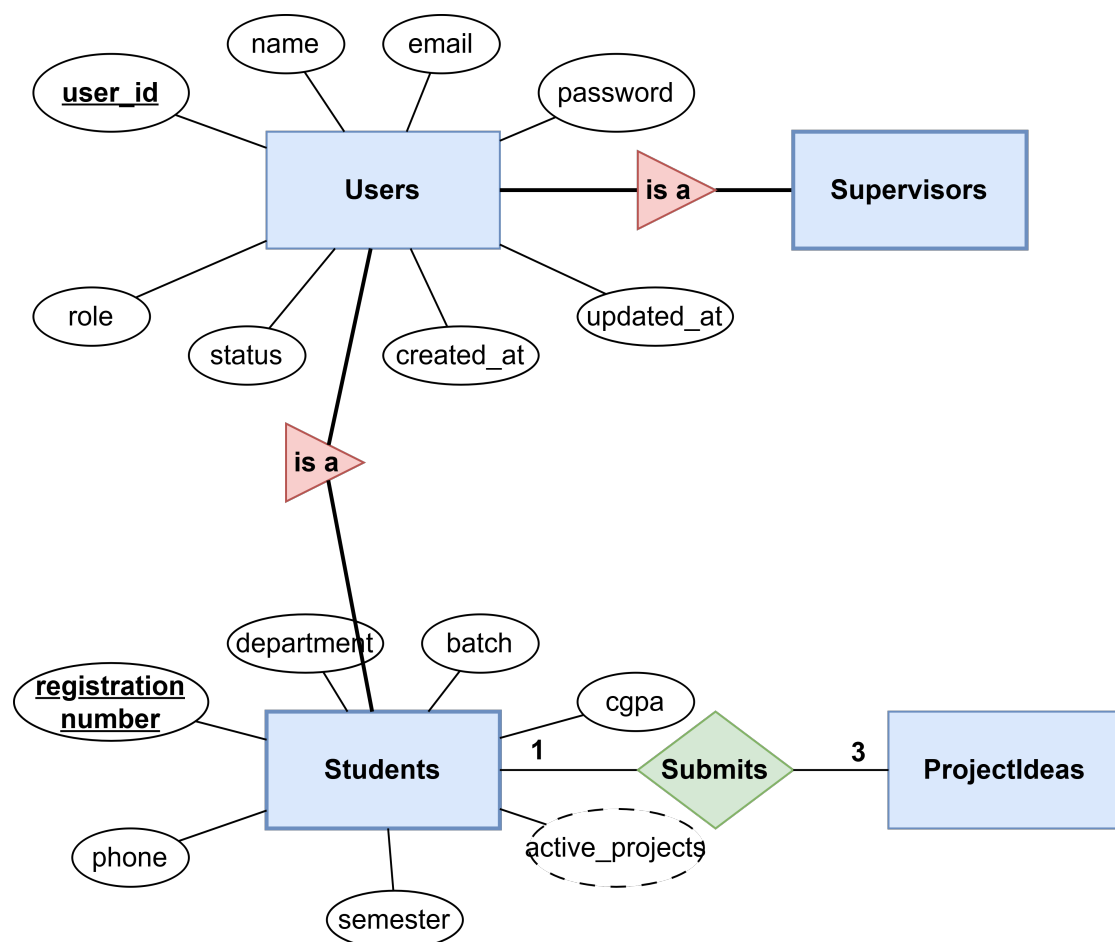


Figure 5.1: Entity-Relationship Diagram part_1 (Logical Model)

Figure 5.1 shows the first part of our database design. You can see how users, roles, and supervisors are connected to each other.



Figure 5.2: Entity-Relationship Diagram part_2 (Logical Model)

Figure 5.2 continues the database structure, focusing on projects and scope documents. This part shows how project ideas flow through the system from submission to approval.

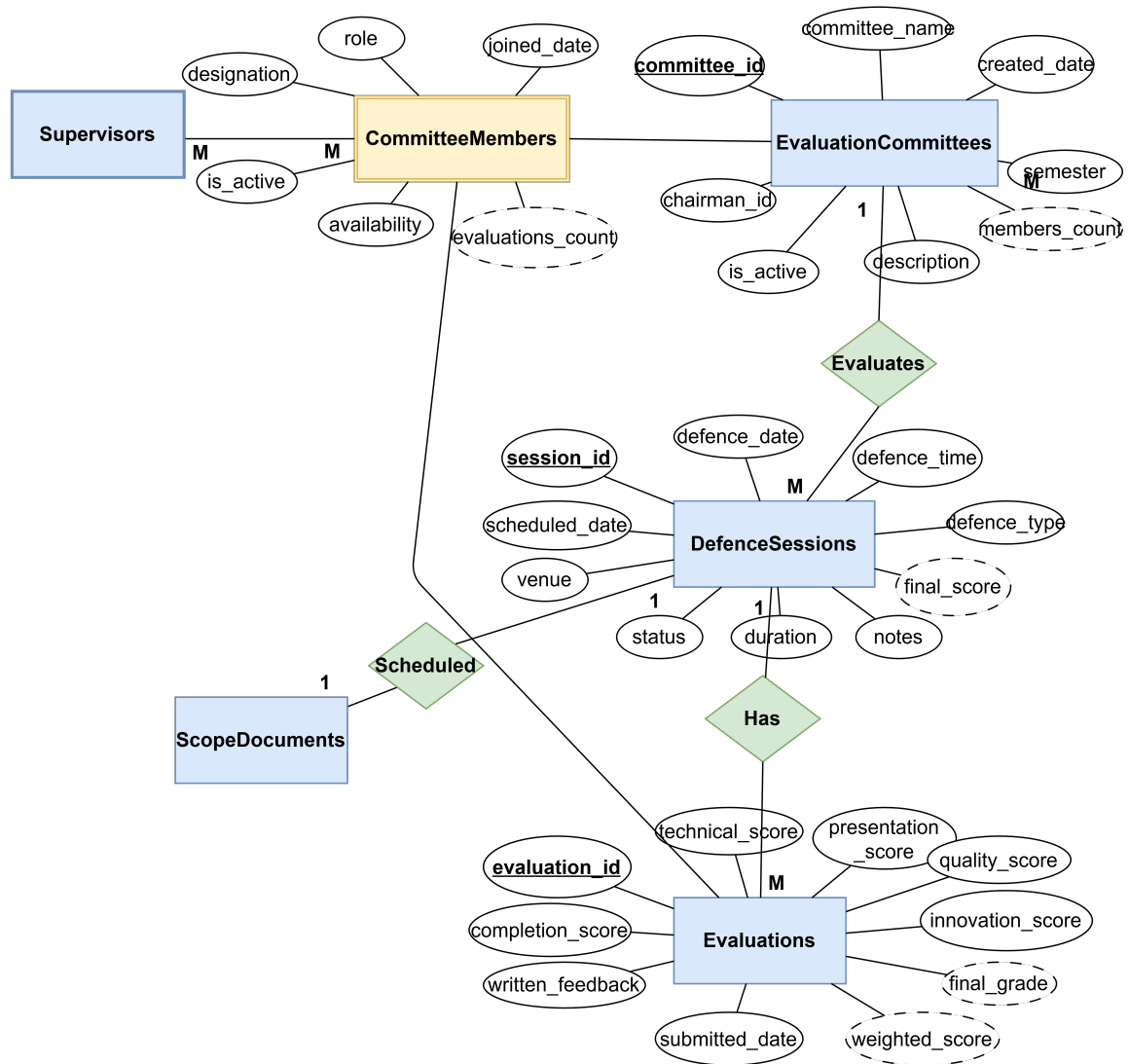


Figure 5.3: Entity-Relationship Diagram part_3 (Logical Model)

Figure 5.3 completes our database design by showing committees, defence sessions, and evaluations. This is where the final evaluation process is structured.

5.1.3 Physical Schema (Selected Tables)

Table 5.1: Physical Schema (Selected Tables)

Table	Columns (Type, Constraints)	PK	FK	Indexes / Notes
users	id BIGINT, name VARCHAR(150), email VARCHAR(190) UNIQUE, password VARCHAR(255), status ENUM('active', 'inactive') DEFAULT 'active', email_verified_at DATETIME NULL, created_at, updated_at, deleted_at NULL	id	—	UNQ(email); IDX(status)
roles	id BIGINT, name VARCHAR(60) UNIQUE, description VARCHAR(255) NULL, created_at, updated_at	id	—	UNQ(name)
role_user (pivot)	user_id BIGINT, role_id BIGINT, assigned_at DATETIME, PRIMARY KEY(user_id, role_id)	(user_id, role_id)	user_id → users.id; role_id → roles.id	IDX(role_id)
supervisors	id BIGINT, user_id BIGINT UNIQUE, capacity TINYINT DEFAULT 6, interests TEXT, office VARCHAR(80) NULL, created_at, updated_at	id	user_id → users.id (ON DELETE CASCADE)	IDX(capacity)

(continued from previous page)

Table	Columns (Type, Constraints)	PK	FK	Indexes / Notes
projects	id BIGINT, student_id BIGINT, supervisor_id BIGINT, title VARCHAR(200), description TEXT, status ENUM('draft', 'pending', 'revision', 'approved', 'rejected') DEFAULT 'pending', submitted_at DATETIME NULL, approved_at DATETIME NULL, created_at, updated_at, deleted_at NULL		student_id → users.id; supervisor_id → supervisors.id	→ IDX(supervisor_id, status); → UNQ(student_id, title)
scope_documents	id BIGINT, project_id BIGINT, version INT, file_path VARCHAR(255), notes TEXT NULL, status ENUM('pending', 'approved', 'rejected') DEFAULT 'pending', submitted_at DATETIME, reviewed_at DATETIME NULL, created_at, updated_at		project_id → projects.id DELETE CASCADE)	→ UNQ(project_id, version); CAS- IDX(status)
committees	id BIGINT, name VARCHAR(120), created_by BIGINT, created_at, updated_at		created_by → users.id	→ —
committee_members	committee_id BIGINT, user_id BIGINT, role ENUM('member', 'chair') DEFAULT 'member', user_id PRIMARY KEY(committee_id, user_id)		committee_id → committees.id; user_id → users.id	→ IDX(role)

(continued from previous page)

Table	Columns (Type, Constraints)	PK	FK	Indexes / Notes
defence_sessions	id BIGINT, project_id BIGINT, committee_id id BIGINT, venue VARCHAR(120), starts_at DATETIME, ends_at DATETIME, status ENUM('scheduled', 'completed', 'cancelled') DEFAULT 'scheduled', created_at, updated_at		project_id projects.id; committee_id committees.id	→ IDX(committee_id, starts_at); → CHK(ends_at > starts_at)
evaluations	id BIGINT, defence_session_id BIGINT, evaluator_id BIGINT, rubric JSON, score DECIMAL(5,2), feedback TEXT, submitted_at DATETIME, created_at, updated_at		defence_session_id → defence_sessions.id; evaluator_id → users.id	UNQ(defence_session_id, evaluator_id)

Table 5.1 shows the actual database table structure with all the main tables and their relationships. You can see how everything connects together properly.

5.1.4 Referential and Business Constraints

- A Student may have at most 3 concurrent project *ideas* in non-final statuses (draft, pending, revision).
- A Supervisor must not exceed their **capacity** of active approved projects (default 6, hard cap 8).
- A Defence Session must be scheduled at least 7 days in advance; a Supervisor cannot be a member of the same project's committee.
- Each Evaluation must be submitted by assigned committee members only, within 24 hours post-session.

5.2 Deployment Requirements

5.2.1 Hardware Sizing (Baseline)

Table 5.2: Hardware Sizing (Baseline)

Component	Spec	Notes
App Server	2 vCPU, 4 GB RAM, 40 GB SSD	PHP-FPM, Nginx/Apache; scale horizontally for peak
DB Server	4 vCPU, 8 GB RAM, 100 GB SSD	MySQL 8; automated backups; point-in-time recovery
Cache/Queue	Managed Redis (cache + queues)	Separate from DB; improves throughput and UX
Storage	Block/Object storage	For document uploads (PDF scope docs)

Table 5.2 shows what kind of hardware we need to run this system. The specs are pretty reasonable for most university environments.

5.2.2 Software Stack

- OS: Ubuntu LTS; Web: Nginx 1.24+

- Runtime: PHP 8.3+, Laravel 12.x, Composer 2.x, Node.js 20+ (assets), Redis 7.x
- Database: MySQL 8.0+ compatibility

5.2.3 Environment Variables (Required)

Table 5.3: Environment Variables (Required)

Variable	Purpose
APP_ENV, APP_URL, APP_KEY	Application mode/URL and encryption key (base64)
DB_HOST, DB_PORT, DB_DATABASE, DB_USERNAME, DB_PASSWORD	Database connectivity (TLS recommended)
CACHE_DRIVER, SESSION_DRIVER	Prefer <code>redis</code> in staging/prod; <code>file</code> in dev
QUEUE_CONNECTION	<code>redis</code> or <code>database</code>
FILESYSTEM_DISK, FILE_MAX_MB	Upload disk (local/s3) and size policy
LOG_CHANNEL, LOG_LEVEL	Centralized logging and verbosity

Table 5.3 lists all the configuration settings you need to set up when deploying the system. These tell the app how to connect to everything.

5.2.4 Network & Security

- Enforce HTTPS (TLS 1.2+), and modern ciphers; redirect HTTP→HTTPS.
- Open ports: 443 (HTTPS), 80 (redirect), 22 (SSH restricted), 3306 (DB internal only), 6379 (Redis internal).

5.3 Graphic User Interfaces (GUIs)

5.3.1 Screens (Spec Summary)

Table 5.4: Screens (Spec Summary)

View	Purpose	Key Fields / Components (Validation)
Login	Authenticate user	Email (required, email); Password (required, min length); Remember me; Forgot password link
Student Dashboard	Overview and status	Cards: Idea count, Pending reviews, Scope status; List: recent activity; CTA buttons
Submit Project Idea	Capture idea details	Title (required, unique per student); Description (required); Objectives (list); Supervisor (required, from available slots)
Supervisor Dashboard	Review workloads	Pending ideas table; Capacity bar; Filters by status/date; Approve/Reject/Request Changes actions
Scope Upload	Upload PDF scope	File (PDF only, max 10MB); Version notes (optional); Submit (validates type/size)
Committee Management	Build committees	Committee name; Member selector (min 2); Roles (chair/member); Save
Defence Scheduler	Schedule sessions	Project picker (approved only); Committee; Venue; Start/End (conflict checks); Create schedule
Evaluation Form	Score & feedback	Rubric criteria (required scores); Overall feedback (required); Submit/Save draft
Admin Reports	KPIs and exports	Filters (date, status, supervisor); Charts (counts/throughput); Export CSV/XLS/PDF

Table 5.4 breaks down all the main screens and their features. This gives you a quick overview of the user interface.

5.3.2 User Interface Flow Diagram

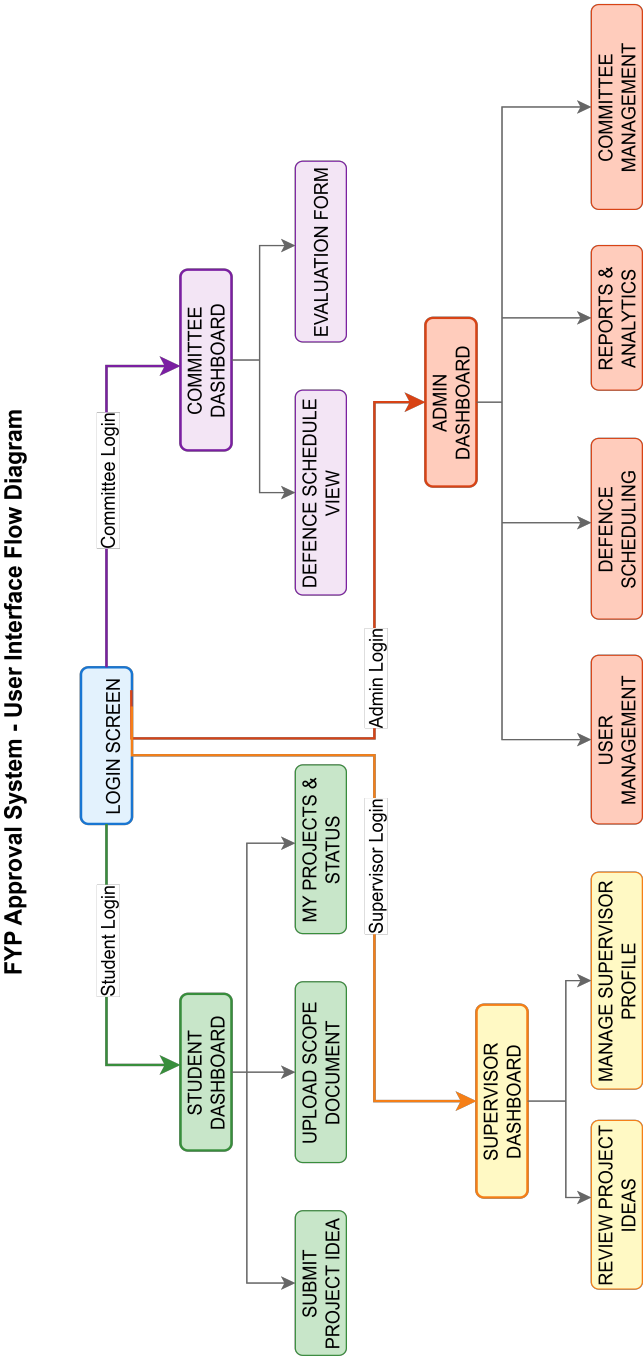
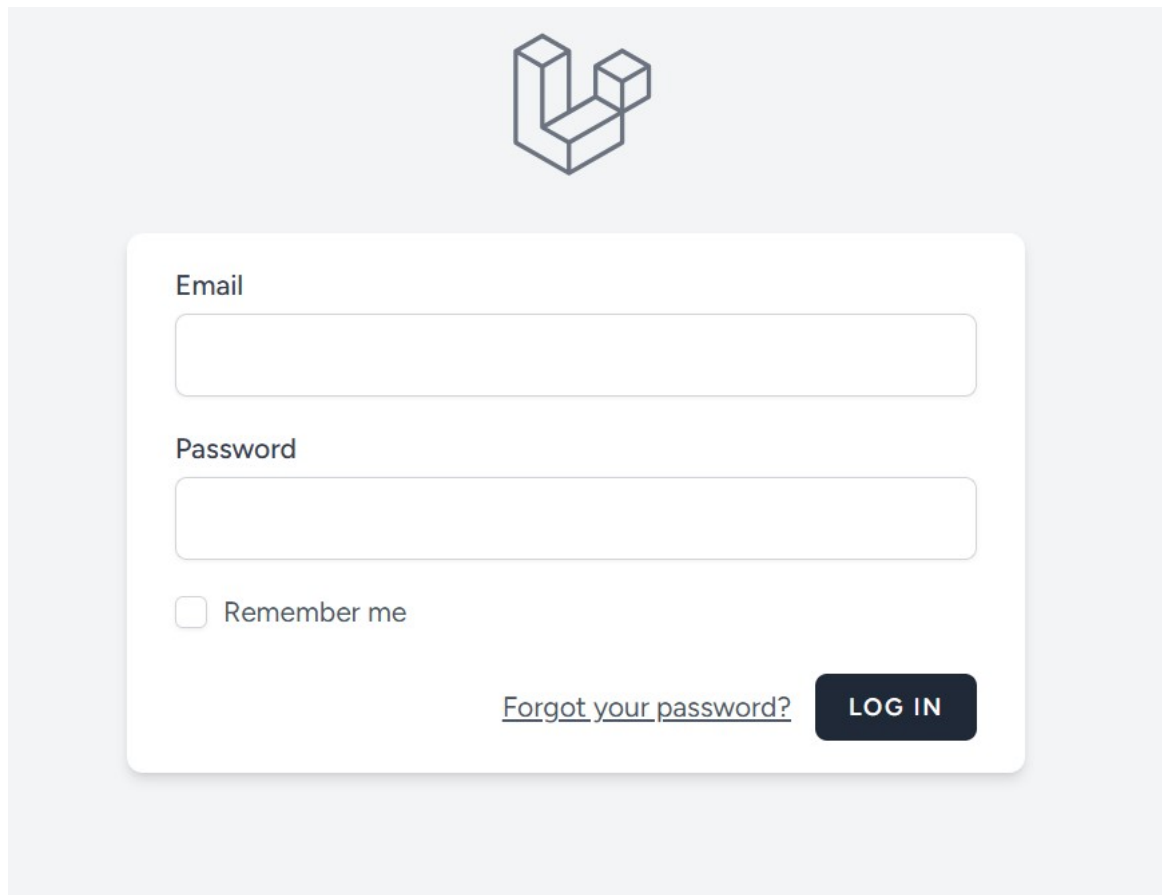


Figure 5.4: User Interface Flow Diagram

Figure 5.4 shows the flow of the user interface screens.

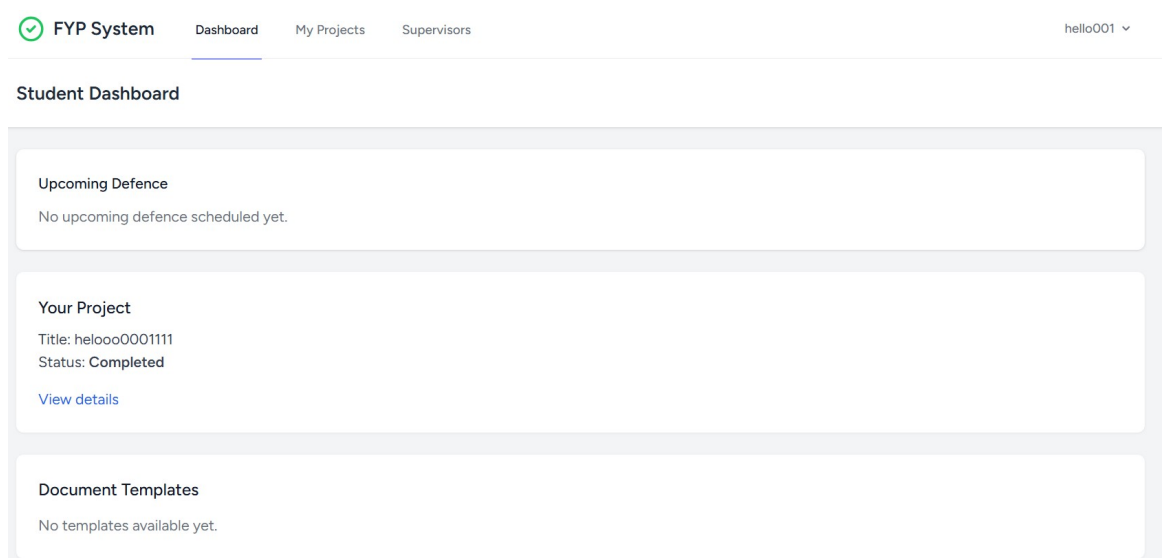
5.3.3 Representative UI Designs



The login screen features a light gray background. At the top center is a logo consisting of three stacked cubes. Below the logo is a white rounded rectangle containing the login form. The form has two input fields: 'Email' and 'Password'. Below the 'Password' field is a checkbox labeled 'Remember me'. At the bottom right of the form is a dark blue 'LOG IN' button. To the left of the button is a link that says 'Forgot your password?'.

Figure 5.5: Login Screen

Figure 5.5 shows our straightforward login screen where users enter their credentials.



The student dashboard has a light gray background. At the top is a navigation bar with a green checkmark icon, the text 'FYP System', and three links: 'Dashboard', 'My Projects', and 'Supervisors'. The 'Dashboard' link is underlined. On the right side of the navigation bar is a user profile section with the text 'hello001' and a dropdown arrow. Below the navigation bar is a section titled 'Student Dashboard'. This section contains three white rounded rectangles. The first rectangle is titled 'Upcoming Defence' and contains the text 'No upcoming defence scheduled yet.' The second rectangle is titled 'Your Project' and contains the text 'Title: helooo0001111' and 'Status: Completed'. Below this text is a link that says 'View details'. The third rectangle is titled 'Document Templates' and contains the text 'No templates available yet.'

Figure 5.6: Student Dashboard

Figure 5.6 is the student dashboard where they can see project status and navigate the system.

The screenshot shows the 'Submit Project Idea' form in the FYP System. The header includes the system name 'FYP System' with a green checkmark icon, and navigation links for 'Dashboard', 'My Projects' (which is underlined), and 'Supervisors'. A user profile 'hello001' with a dropdown arrow is in the top right. The form title 'Submit Project Idea' is at the top left of the main content area. The form itself is a light gray box containing three input fields: 'Project Title' with a placeholder 'e.g., AI-Powered Healthcare Chatbot', 'Description' with a placeholder 'Provide a brief overview of your project.', and 'Select Supervisor' with a placeholder 'Select a supervisor' and a dropdown arrow. At the bottom right of the form are two buttons: 'Save as Draft' (gray) and 'Submit for Approval' (blue).

Figure 5.7: Idea Submission Screen

Figure 5.7 shows the project idea submission form with all the required fields.

The screenshot shows the 'Upload Scope Document' form in the FYP System. The header is identical to Figure 5.7. The form title 'Upload Scope Document' is at the top left of the main content area. The form is a light gray box. It starts with 'Project: 1234ki' followed by a note: 'Please upload your final scope document. The document must be in PDF format and no larger than 10MB.' Below this is the label 'Scope Document (PDF only):'. The main input area contains a 'Choose File' button and the text 'No file chosen'. At the bottom right is a blue 'Upload Document' button.

Figure 5.8: Scope Document Upload Screen

Figure 5.8 is where students upload scope documents with file validation.

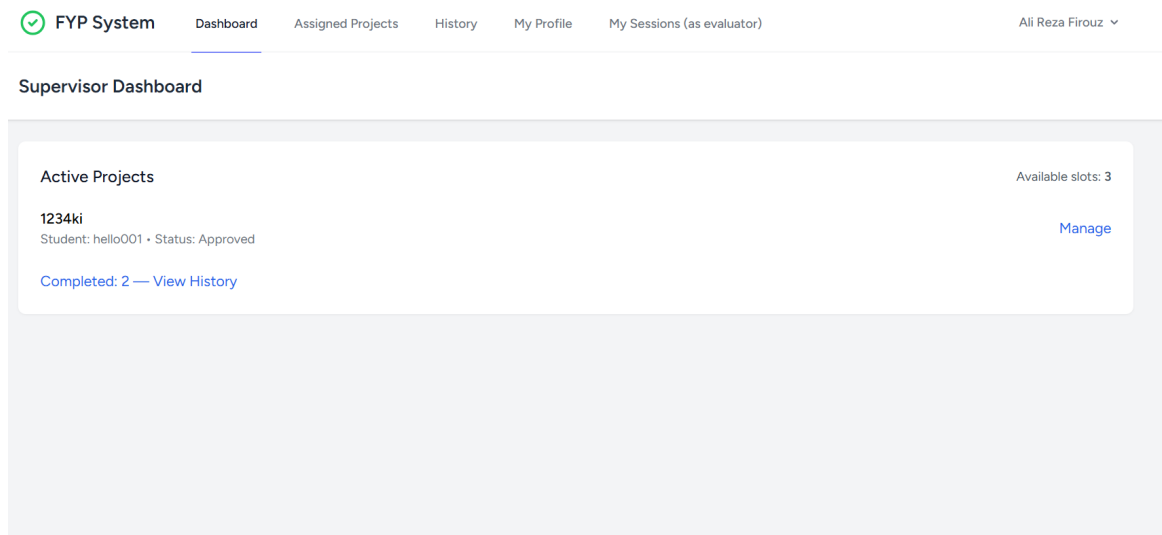


Figure 5.9: Supervisor Dashboard

Figure 5.9 shows the supervisor dashboard with capacity tracking and pending reviews.

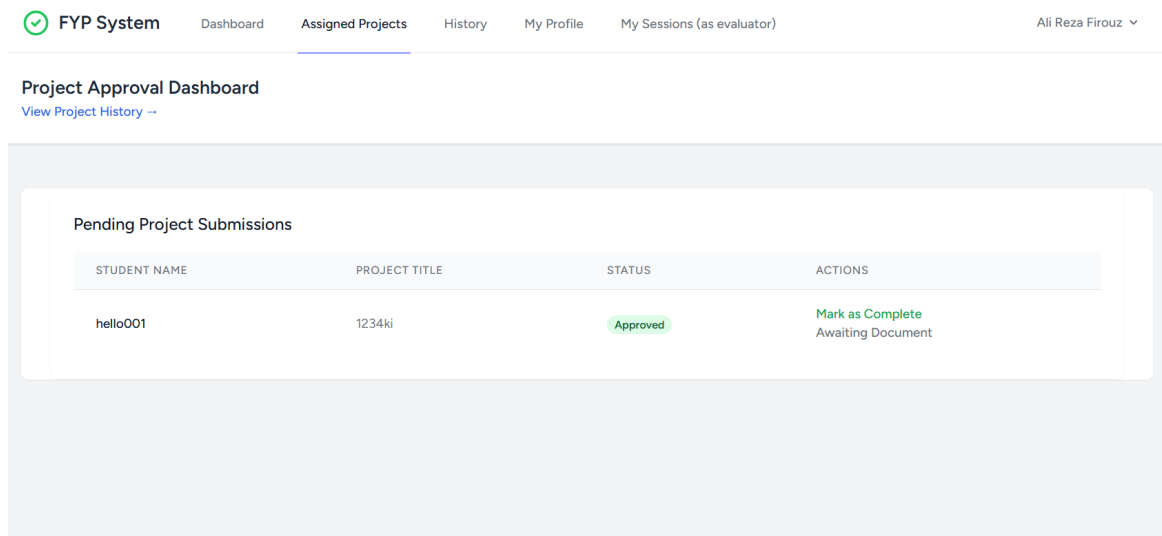


Figure 5.10: Supervisor Review Screen

Figure 5.10 is where supervisors review and approve project ideas.

Members (Supervisors only)

Available Evaluators

— Select supervisor —

▼

Committee Role

Member

▼

Add

Ali Reza Firouz

ali@gmail.com — Member

Evaluator (Supervisor)

Assigned

Remove

Diana Miller

diana.m@example.com — Member

Evaluator (Supervisor)

Assigned

Remove

Schedule Defence Session

All Sessions

Evaluator (supervisor) added to committee.

Project

1234ki

▼

Scheduled At

dd/mm/yyyy --:-- --

📅

Venue

e.g., Seminar Hall A

Schedule

Figure 5.11: Defence Schedule Screen

Figure 5.11 shows the defence scheduling interface with automatic conflict checking.

✓

FYP System

Dashboard

Projects

Templates

Users

Committees

Evaluators

mona ▼

Evaluation Committees

New Committee

00909090

0 members

Manage

hellooo01

0 members

Manage

committee

1 members

Manage

Figure 5.12: Committee Management Screen

Figure 5.12 is where administrators create and manage evaluation committees.

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5.4 Algorithms / Flowcharts

5.4.1 Scheduling Conflict Detection

Objective: Prevent overlapping sessions for venues and committee members.

- Given candidate [starts_at, ends_at], venue V, and committee C:
- Check: no defence_sessions s.t. same V and time windows overlap.
- Check: for each member in C, no assigned session overlaps in the window.
- Enforce buffer (e.g., 30 minutes) between sessions at same venue.

Pseudocode (verbatim):

```
function isSchedulable(projectId, committeeId, venue, startsAt, endsAt):
  if exists session where venue = venue AND overlap(startsAt, endsAt):
    return false, "Venue conflict"
  for member in committeeMembers(committeeId):
    if exists session where member assigned AND overlap(startsAt, endsAt):
      return false, "Member conflict: " + member.name
  if (durationBetween(lastSessionAtVenueEnd, startsAt) < 30min):
    return false, "Buffer too small"
  return true, "OK"
```

5.4.2 Supervisor Capacity Enforcement

Rule: active_approved_projects $\leq$ supervisors. capacity (hard cap 8).

```
function canApprove(supervisorId):
  active = countApprovedProjects(supervisorId)
  cap = getCapacity(supervisorId) // default 6, max 8
  return active < cap
```

5.4.3 Evaluation Aggregation

Goal: Compute final score when all committee evaluations submitted.

```
function finalizeIfComplete(defenceSessionId):
  expected = countCommitteeMembers(defenceSessionId)
  submitted = countEvaluations(defenceSessionId)
  if submitted == expected:
    scores = getScores(defenceSessionId)
```

```

final = average(scores)
storeFinal(defenceSessionId, final)
updateProjectStatus(defenceSessionId, final)

```

5.4.4 CSV Bulk Import Validation

Goal: Ensure atomic user creation from CSV (all-or-nothing).

```

function importUsers(csv):
    beginTransaction()
    try:
        for row in csv:
            validate(row. email unique, password policy, role exists)
            createUser(row)
        commit()
    except ValidationError as e:
        rollback()
        report(e)

```

5.4.5 Flowchart Placeholders

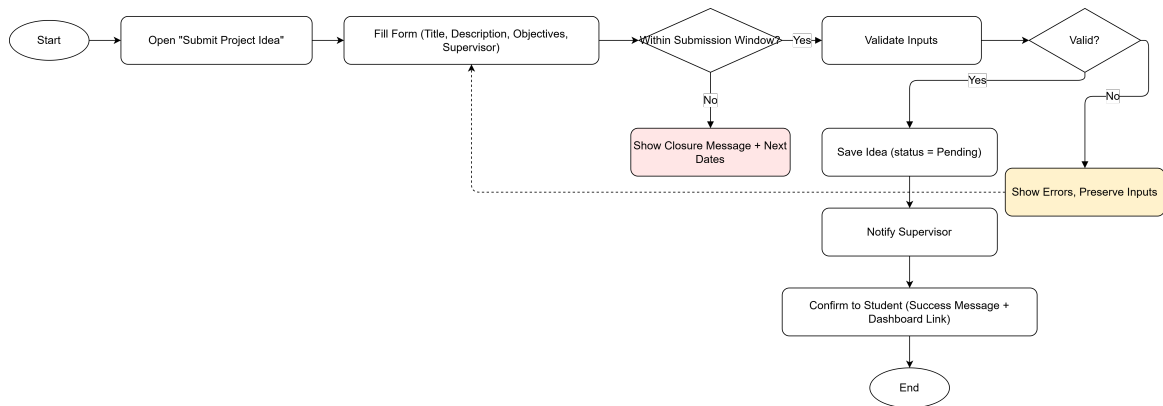


Figure 5.13: Flowchart: Idea Review and Approval

Figure 5.13 shows the complete workflow for project idea processing. You can see all the decision points and different paths.

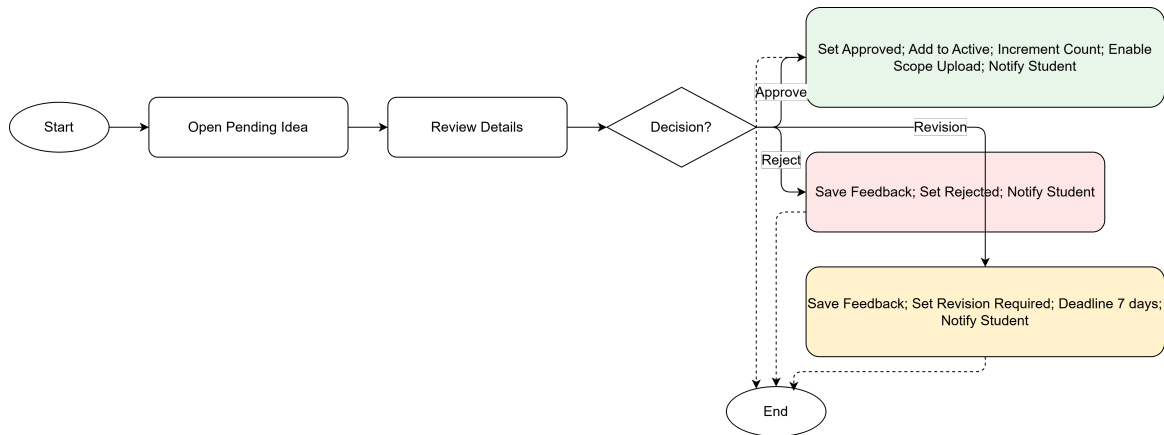


Figure 5.14: Flowchart: Idea Review by Supervisor

Figure 5.14 focuses on the supervisor review process and the different outcomes possible.

Implementation Notes

- Database migrations encode the schema and constraints; seeders provide sample roles and users.
- Feature flags may be used to roll out new rubric criteria without downtime.

6 Conclusion

This Software Requirements Specification document has presented a comprehensive overview of the FYP Proposal Approval System for COMSATS University Islamabad, Vehari Campus. The document has detailed:

Requirements Coverage

- Functional requirements across five major modules (User Management, Supervisor Profile Management, Project Idea and Approval, Scope Document Management, and Evaluation Committee Management)
- Non-functional requirements addressing usability, performance, security, reliability, maintainability, and portability
- Clear use case descriptions with normal flows, alternative flows, and exception handling

Design Specifications

- Data flow diagrams showing system processes and information flow
- Interaction diagrams illustrating component collaboration
- Entity-Relationship Diagram defining the database schema

Success Criteria

The project will be considered successful when:

- All high-priority functional requirements are implemented and tested
- Non-functional requirements meet specified thresholds
- User acceptance testing is completed with stakeholder approval

Acknowledgments

This project would not be possible without the support and guidance of:

- Project supervisor for technical guidance
- FYP committee for providing domain knowledge and requirements
- Students and faculty who participated in requirements gathering
- COMSATS University Islamabad, Vehari Campus, for facilitating this initiative

Bibliography

- [1] COMSATS University Islamabad. (2024). Final year project guidelines and procedures. Department of Computer Science, COMSATS University Islamabad, Vehari Campus.
- [2] Dennis, A., Wixom, B. H., & Tegarden, D. (2015). *Systems analysis and design: An object-oriented approach with UML* (5th ed.). John Wiley & Sons.
- [3] International Organization for Standardization. (2013). *ISO/IEC 27001:2013 - Information technology — Security techniques — Information security management systems — Requirements*. ISO/IEC.
- [4] Laravel LLC. (2024). Laravel 11.x documentation. <https://laravel.com/docs/12.x>
- [5] MySQL Documentation Team. (2024). *MySQL 8.0 reference manual*. Oracle Corporation. <https://dev.mysql.com/doc/refman/8.0/en/>
- [6] National Institute of Standards and Technology. (2018). *NIST cybersecurity framework version 1.1*. U.S. Department of Commerce.
- [7] Object Management Group. (2017). *OMG unified modeling language (UML) version 2.5.1*. <https://www.omg.org/spec/UML/2.5.1/>
- [8] OWASP Foundation. (2021). *OWASP top ten web application security risks 2021*. <https://owasp.org/www-project-top-ten/>
- [9] PHP Group. (2024). *PHP 8.3 manual*. <https://www.php.net/manual/en/>
- [10] Tailwind Labs Inc. (2024). *Tailwind CSS documentation*. <https://tailwindcss.com/docs>
- [11] Wieggers, K. E., & Beatty, J. (2013). *Software requirements* (3rd ed.). Microsoft Press.

A Appendix

A.1 Glossary

Table A.1: Glossary of Terms

Term	Definition
FYP	Final Year Project — Capstone project required for graduation.
Scope Document	Formal document detailing project objectives, deliverables, and timeline.
Defence Session	Oral presentation and evaluation of the project by the committee.
Evaluation Committee	Panel of faculty members assigned to assess projects.
RBAC	Role-Based Access Control — Security model for managing permissions.
MVC	Model–View–Controller — Architectural pattern for web applications.
CRUD	Create, Read, Update, Delete — Basic database operations.
SRS	Software Requirements Specification.
UAT	User Acceptance Testing.
Rubric	Standardized evaluation criteria used by committee members to score projects.
Dashboard	Custom home screen showing relevant information based on user role.

Table [A.1](#) provides definitions of key terms used in the documentation.

A.2 Acronyms

Table A.2: List of Acronyms

Acronym	Full Form
API	Application Programming Interface.
CSRF	Cross-Site Request Forgery.
CSV	Comma-Separated Values.
CUI	COMSATS University Islamabad.
DFD	Data Flow Diagram.
ERD	Entity-Relationship Diagram.
HTTPS	Hypertext Transfer Protocol Secure.
PDF	Portable Document Format.
SQL	Structured Query Language.
TLS	Transport Layer Security.
UI	User Interface.
URL	Uniform Resource Locator.
XSS	Cross-Site Scripting.
JSON	JavaScript Object Notation.
RTL	Real-time Logging.

Table [A.2](#) lists all technical acronyms used in the document.

A.3 Project Gantt Chart

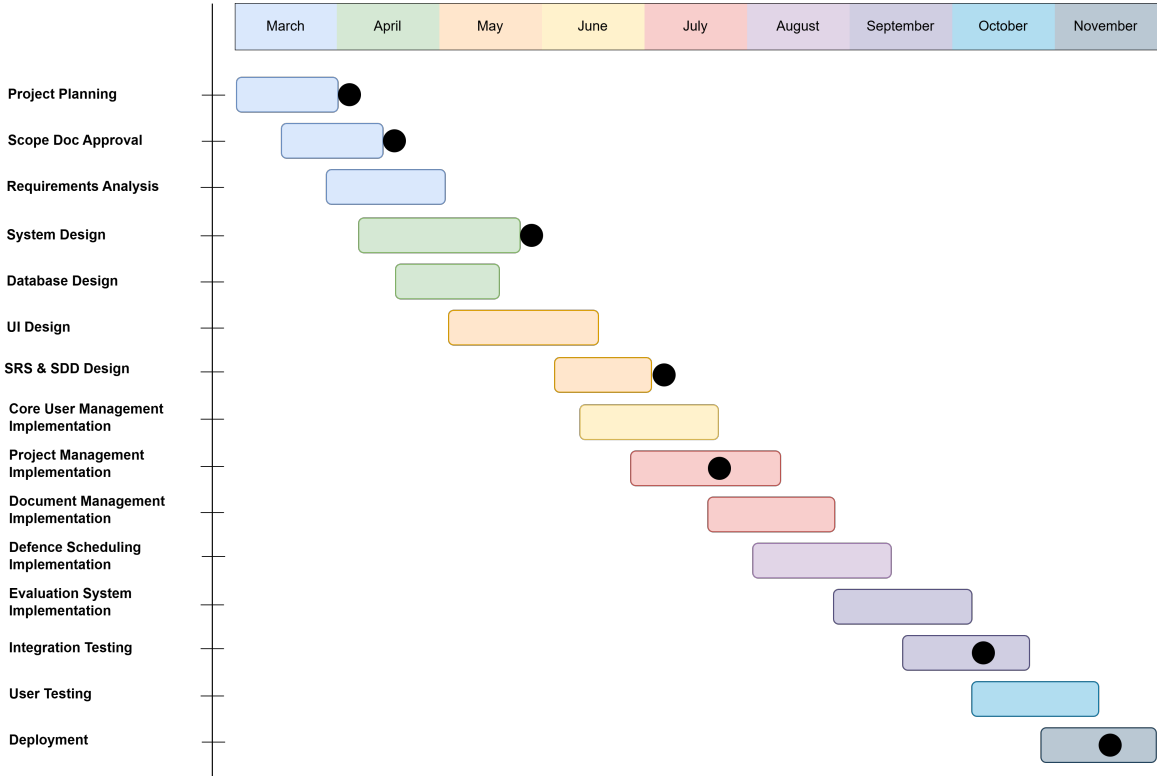


Figure A.1: Gantt Chart

Figure [A.1](#) shows the project timeline.

A.4 User Interface Mockups

Login Page

The mockup shows a login interface for the 'FYP Proposal Approval System'. It features a light blue header with the system name, a central 'Login' title, two input fields for 'Username/Email' and 'Password', a 'Remember me' checkbox, a 'Forgot Password?' link, and a blue 'Login' button.

FYP Proposal Approval System

Login

Username/Email

Password

☐ Remember me [Forgot Password?](#)

Login

Figure A.2: Login Page Mockup

Figure A.2 shows the login interface.

Student Dashboard

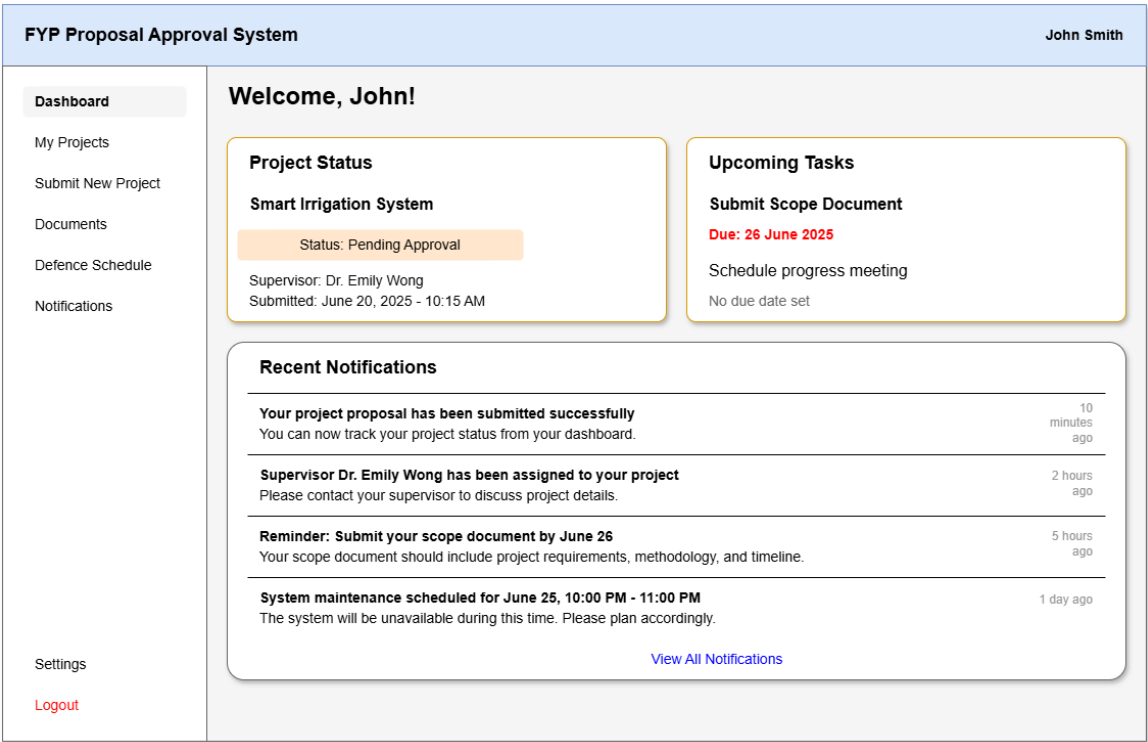


Figure A.3: Student Dashboard Mockup

Figure A.3 shows the student dashboard displaying project status, pending reviews, and quick action buttons.

Project Idea Submission Form

Submit New Project Proposal

John Smith

Project Details

References

Review & Submit

Dashboard

My Projects

Submit New Project

Documents

Defence Schedule

Notifications

Settings

Logout

Project Title *

Smart Irrigation System using IoT and AI

Project Description *

Preferred Supervisor *

Dr. Emily Wong

Department *

Computer Science

Back

Next

Figure A.4: Idea Submission Form Mockup

Figure A.4 shows the project idea submission form with fields for title, description, objectives, and supervisor selection.

Supervisor Review Interface

FYP Proposal Approval System

Dr. Emily Wong

Dashboard

Pending Reviews

Approved Projects

Document Reviews

Defence Schedule

Notifications

Settings

Logout

Project Review: Smart Irrigation System using IoT and AI

Project Details

Title: Smart Irrigation System using IoT and AI
Student: John Smith (CS21045)
Submitted: June 20, 2025 - 14:30
Description: This project aims to develop an automated irrigation system that uses IoT sensors and AI algorithms to optimize water usage for agricultural applications. The system will include soil moisture sensors, temperature sensors, and a valve control system that can be monitored and controlled remotely via a web and mobile application.

Feedback

Enter your feedback here...

Decision

Approve

Request Revision

Reject

Back to Reviews

Figure A.5: Supervisor Review Form Mockup

Figure A.5 shows the supervisor interface for reviewing project ideas with options to approve, reject, or request revisions.

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Defence Scheduling Interface

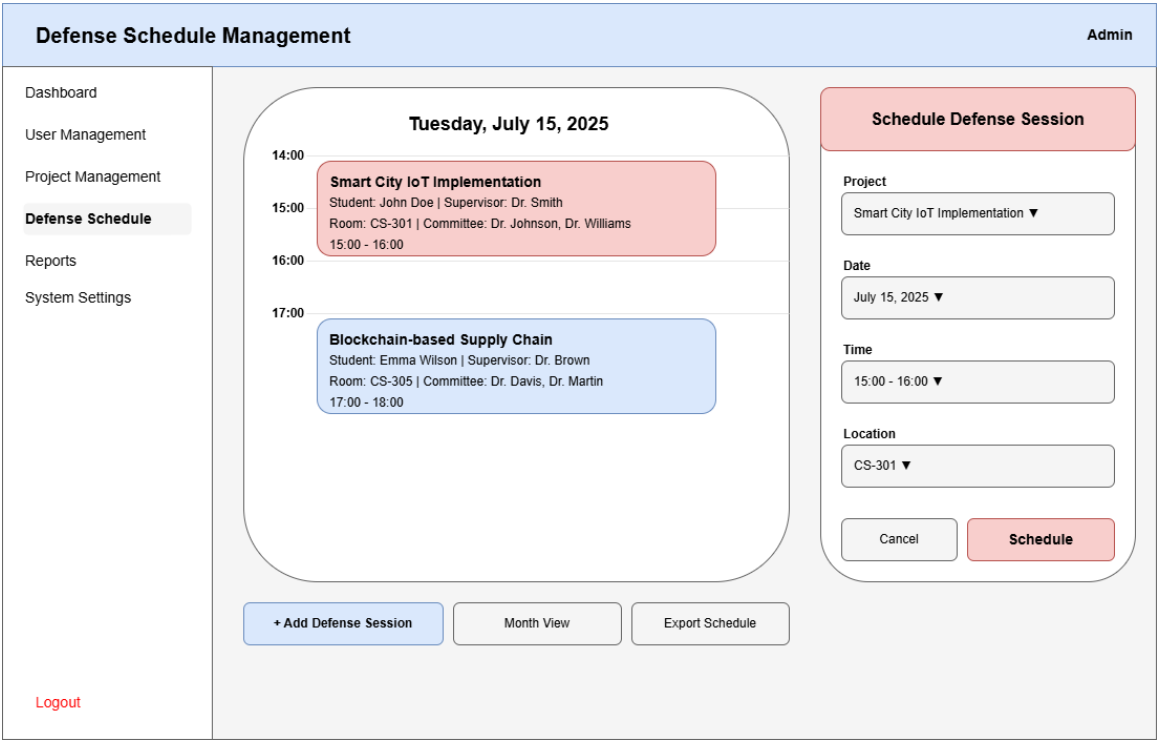


Figure A.6: Defence Scheduling Interface Mockup

Figure A.6 shows the administrative interface for scheduling defence sessions with project, committee, venue, and time selection.

Administrative Reports Dashboard

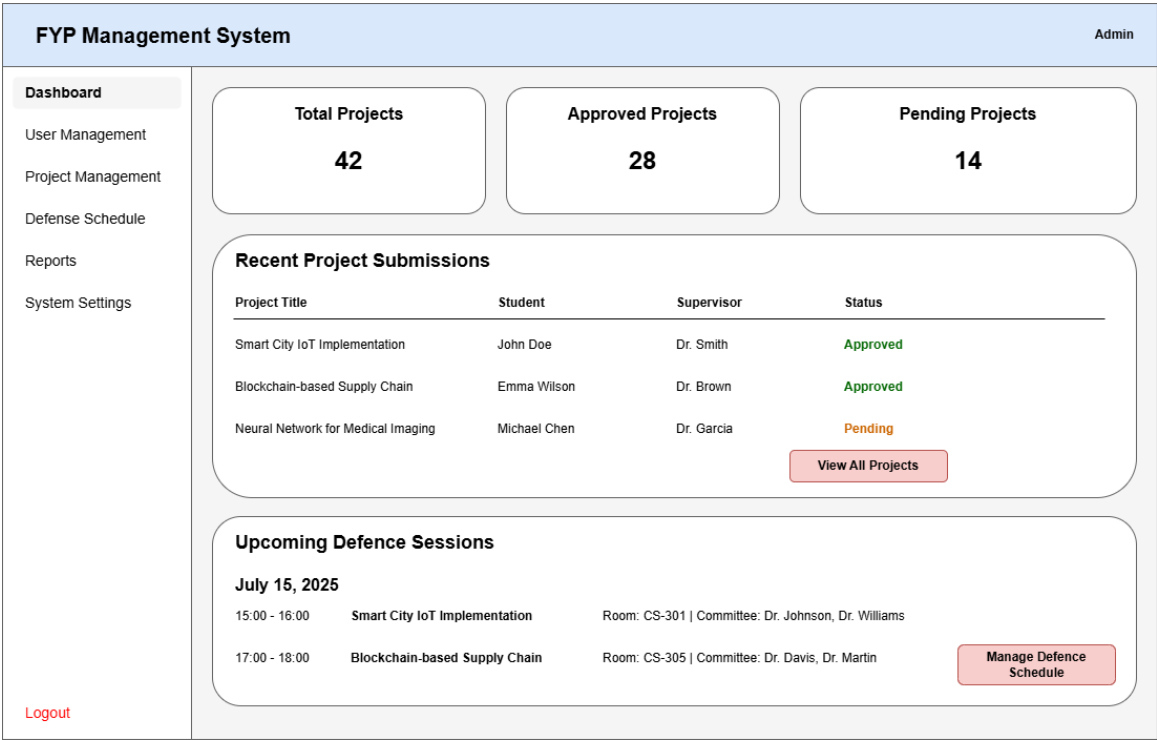


Figure A.7: Admin Dashboard Mockup

Figure A.7 shows the administrative dashboard displaying system statistics, charts, and report generation tools.